



part IV

to prove its worth

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business decisions

With new statistical theories cropping up almost daily during the 1960s, the paltry number of practical applications in the public arena was becoming a professional embarrassment.

Harvard's John W. Pratt complained that Bayesians and frequentists alike were publishing "too many minor advances extracted from real or mythical problems, sanitized, rigorized, polished, and presented in pristine mathematical form and multiple forums."¹

Bayesians in particular seemed unwilling to apply their theories to real problems. Savage's rabbit ears and 20-pound chairs were textbook showpieces, even less substantial than Egon Pearson's chestnut foals and pipe-smoking males of 30 years before. They were "dumb," a Bayesian at Harvard Business School protested later;² they lacked the practical world's ring of truth. When analyzing substantial amounts of data even diehard Bayesians preferred frequency. Lindley, already Britain's leading Bayesian, presented a paper about grading examinations to the Harvard Business School in 1961 without mentioning Bayes; only later did he analyze a similar problem using wine statistics and Bayesian methods.

Mathematically and philosophically, the rule was simplicity itself. "You can't have an opinion afterwards," said Pratt, "without having had an opinion before and then updating it using the information."³ But making a belief quantitative and precise—there was the rub.

Until their logically appealing system could prove itself as a powerful problem solver in the workaday world of statistics, Bayesians were doomed to minority status. But who would undertake a cumbersome, complex, techni-

cally fearsome set of calculations to explore a method that was professionally almost taboo? At the dawn of the electronic age, powerful computers were few and far between, and packaged software nonexistent. Bayesian techniques for dealing with practical problems and computers barely existed. Brawn—lots of it—would have to substitute for computer time. The faint of heart need not apply.

Nevertheless, a few hardy, energetic, and supremely inventive investigators tried to put Bayes to work for business decision makers, social scientists, and newscasters. Their exploits dramatize the formidable obstacles that faced anyone trying to use Bayes' rule.

The first to try their luck with Bayes was an unlikely duo at Harvard Business School, Robert Osher Schlaifer and Howard Raiffa. They were polar opposites. Schlaifer was the school's statistical expert, but, in a sign of the times, he had taken only one mathematics course in his life and was an authority on ancient Greek slavery and modern airplane engines. Raiffa was a mathematical sophisticate who became the legendary Mr. Decision Tree, an advisor to presidents and a builder of East-West rapprochement. Together they tackled the problem of turning Bayes into a tool for business decision makers.

Fortunately, Schlaifer enjoyed using his laserlike mind, hyperlogic, and outsider status to bludgeon through convention and orthodoxy. Years later, when Raiffa was asked to describe his colleague, he did so in two words, "imperious and hierarchical."

Schlaifer was an opinionated perfectionist. Plunging into a topic, he saw nothing else. When his passion turned to bicycle racks, he persuaded a Harvard dean to install his new design on campus. Because he loved old engines, an MIT physicist volunteered to come each week to maintain his Model A Ford and hi-fi to his exacting standards. And when he studied consumer behavior, his faculty colleagues pondered instant coffee as soberly as they would have nuclear fusion. Like an autocrat atop an empire, Schlaifer bestowed nicknames on his colleagues: "Uncle Howard" for Raiffa, "Great Man Pratt" for John W. Pratt, and, most memorably, "Arturchick" for his graduate student and near namesake, Arthur Schleifer Jr. All three survived to become chaired professors at Harvard.

Whatever else, the man had panache and fun. Few Harvard professors invited research assistants for Sunday dinners and bottles of 1938 Clos Vougeot, but Schlaifer did. And few took off an entire month or sabbatical year

to relax in Greece or France, as he did with his French-born wife, Geneviève, whom he publicly referred to as Snuggle Buggle.

Schlaifer was not to the manor born. He was born in Vermillion, South Dakota, in 1914 and grew up near Chicago, in a small town where his father was superintendent of schools. At Amherst College he majored in classics and ancient history and took courses in economics and physics. He enrolled in calculus, his only formal math class, solely to capture a large cash prize for the best student. After graduating Phi Beta Kappa at the age of 19, he studied in Athens at the American School of Classical Studies between 1937 and 1939 and earned a Ph.D. in ancient history at Harvard in 1940. Over the next few years he published several articles about religious cults and slavery in ancient Greece. Schlaifer was a quick study, and he filled in for Harvard historians, economists, and physicists as they left for defense work during the Second World War.

Eventually, Schlaifer was assigned to the university's Underwater Sound Laboratory, where sonar was developed. He and theoretical physicist Edwin Kemble tried to silence submarine torpedo propellers the better to attack German U-boats. Schlaifer understood the scientific issues well enough to solve equations using Marchant or Frieden electromechanical calculators and to turn technical reports into prose. The war gave him a voracious appetite for practical, real-world problems, and he abandoned ancient history.

After the war Schlaifer's physics impressed the Harvard Business School enough to hire him to fulfill a departmental obligation: a study of the aircraft engine industry. He turned the low-status assignment into a triumph, a 600-page classic on aviation history, *Development of Aircraft Engines, Development of Aircraft Fuel*.

Between his war work and the book, Schlaifer acquired a usefully intimidating campus reputation as a physicist, later cemented by his obituary in the *New York Times*. He was teaching accounting and production when the business school, despite his spectacularly unsuitable training, assigned him to teach statistical quality control. Knowing nothing about statistics, Schlaifer crammed by reading the dominant theoreticians of the day, Fisher, Neyman, and Egon Pearson. Wartime operations research had mathematized problem solving for two common business problems, inventory control and transportation scheduling. But frequentism offered businesses no help when it came to issues like launching a new product or changing a price.

Schlaifer's publications later described modest pleas for help from a newsstand owner unsure of how many copies of the *Daily Racing Form* to

stock, and from a wholesaler worried about allocating his ten delivery trucks between two warehouses. With luck, business owners might make the right decisions. But given the uncertainties involved in even these simple problems, Schlaifer wondered how they could ever hope to *systematically* make the best possible choices. Even if they could collect additional information by sampling or experimentation, would it be worth the cost?

According to frequentists, objective statistics were synonymous with long-run relative frequency, and probabilities were invalid unless based on repeatable observations. Frequentists dealt with a rich abundance of directly relevant data and took samples to test hypotheses and draw inferences about unknowns. The method worked for repetitive, standardized phenomena such as successive grain crops, genetics, gambling, insurance, and statistical mechanics.

But business executives often had to make decisions under conditions of extreme uncertainty, without sample data. As Schlaifer realized, "Under uncertainty, the businessman is forced, in effect, to gamble, . . . hoping that he will win but knowing that he may lose."⁴ Executives needed a way to assess probabilities without the repeated trials required by frequentist methods. Schlaifer said teaching frequentism made him feel like a jackass. It simply did not address the main problem in business: decision making under uncertainty.

Thinking through the problem, Schlaifer wondered how executives could make decisions based on no data. Whatever prior information they had about the demand for their product was obviously better than none. From there, Schlaifer got to the problem of how sample data should be used and how much money should be spent getting it. Updating prior information with sample data got him to Bayes' rule because it could combine subjectively assessed prior probabilities with objectively attained data. It was a fundamental insight that changed his life.

Schlaifer did not have much mathematics. Innocent of the raging philosophical divide between objectivists and subjectivists, Schlaifer threw away his books and reinvented Bayesian decision theory from scratch. As a self-made statistician working in a business school, he owed nothing to the statistical establishment. And given his iconoclastic zeal, he brazenly challenged giants in the field. Like Savage, Schlaifer was combining uncertainty with economics in order to make decisions. Savage thought Schlaifer was "hot as a pistol, sharp as a knife, clear as a bell, quick as a whip, and as exhausting as a marathon run."⁵

Schlaifer realized that his mathematical competence was almost nil, “of order of magnitude epsilon.”⁶ To compensate, he worked 75 to 80 hours a week, puffed through four packs of unfiltered cigarettes a day, and traced his thoughts with different colored chalk across the blackboard in his smoke-filled office. Single-mindedly pursuing what he thought was relevant to the real world, he would lurch into one theory, back off, try it another way, and race on to yet another. His voice boomed down the hall almost hourly—“Oh, my God!” or “How could I have been so dumb?”—as he overturned one firmly held opinion with another. Ever curious, he demanded the absolute, best possible analyses. He realized he needed mathematical help.

Hearing about a young closet Bayesian named Howard Raiffa at Columbia University, Schlaifer scouted him out and persuaded Harvard to hire him. For the next seven years Raiffa and Schlaifer collaborated closely. Raiffa went on to become an international negotiator who wielded enormous influence on education, business, the law, and public policy in the United States and abroad. But he always regarded Schlaifer as “the great man. . . . I revered him; I was in awe of him. . . . He was so positive, so certain, so opinionated, but so smart, so smart. . . . [He was] “a man—a real man—who independently discovered Bayesianism, mocked those who didn’t agree with him, and not only theorized and philosophized but applied the approach to real problems.” Raiffa called him “the—not a but the—most important person in my intellectual development.”⁷

Both Schlaifer and Raiffa were elite intellectuals, but Schlaifer’s style was arrogant while Raiffa was, in the words of a collaborator, “a sweetheart, a very warm, open, embracing person.”⁸ Schlaifer was Phi Beta Kappa in the Ivy League; Raiffa attended City College of New York, “the college of choice for poor and middle-income students in New York—I was on the poor side.”⁹ During the Second World War, a misplaced template for grading the air force’s arithmetic and elementary algebra test flunked Raiffa and doomed the future advisor to U.S. presidents—not to a prestigious research laboratory like the one where Schlaifer spent the war but to three rounds of basic training and to grunt assignments in cooks-and-bakers school, meteorology, and a radar blind-landing system.

Anti-Semitism finally determined Raiffa’s career choice. One day he overheard his army sergeants saying they wanted to line up America’s Jews on a beach and use them for target practice. Later, real estate agents in Fort Lauderdale, Florida, refused to find housing for Raiffa and his wife because they were Jewish. When a friend told him that engineering and science also

discriminated against Jews, Raiffa was prepared to believe it. Then he learned that insurance actuaries were graded on objective, competitive examinations. Seeking a field where competence counted more than religion, Raiffa enrolled in the University of Michigan's actuarial program, where Arthur Bailey had studied.

To Raiffa's great surprise, he became a superb and "deliriously happy" student who raced through a bachelor's degree in mathematics, a master's degree in statistics, and a doctorate in mathematics in six years between 1946 and 1952. "In the year I studied statistics, I don't think I heard the word 'Bayes.' As a way of inference, it was nonexistent. It was all strictly Neyman-Pearson, classical, objectivistic (frequency-based) statistics."¹⁰

Although Schlaifer had embraced Bayes in one fell swoop, Raiffa inched grudgingly toward its subjectivity. But reading John von Neumann and Oskar Morgenstern's book *Game Theory* (1944), he instinctively assessed how others would play in order to determine how he himself should compete: "In my naiveté, without any theory or anything like that. . . . [I began] assessing judgmental probability distributions. I slipped into being a subjectivist without realizing how radically I was behaving. That was the natural thing to do. No big deal."¹¹

When Raiffa gave a series of seminars on Abraham Wald's new book *Statistical Decision Functions*, he discovered it was full of Bayesian decision-making rules for use in a frequentist framework. Independently of Turing and Barnard, Wald had discovered sequential analysis for testing ammunition while working with the Statistical Research Group in the Fire Control Division of the National Defense Research Committee. Though a confirmed frequentist, he sometimes solved problems in a curiously roundabout manner. After inventing a Bayesian prior, he would solve the Bayesian version of his problem and then analyze its frequentist properties. He also said that every good decision procedure is Bayesian and confided to the statistician Hilda von Mises that he was a Bayesian but did not dare say so publicly. His work would have a large impact on many mathematical statisticians and decision theorists, including Raiffa.

Until Wald's book appeared, the word "Bayesian" had referred only to Thomas Bayes' controversial suggestion about equal priors, not to his theorem for solving inverse probability problems. After Wald died in a plane crash in India in 1950, the Columbia University statistics department hired Raiffa to teach Wald's course. Raiffa kept a day ahead of his students by reading the textbook every night. He was moving gradually toward a viewpoint opposed

by almost every statistics department in the country, including Columbia's. Giving up "scientific" objectivity and embracing subjectivity would not be easy.

At first, like Schlaifer, Raiffa taught straight frequentism, using the then-canonical Neyman-Pearson theory, tests of hypotheses, confidence intervals, and unbiased estimation. But by 1955, Raiffa, like Schlaifer, no longer believed these concepts were central. Columbia faculty members were auditing Raiffa's lectures, and his transformation into a closet Bayesian left him a nervous wreck. He did not "come out" because colleagues whom he admired greatly were vociferously opposed to Bayesianism. "Look, Howard, what are you trying to do?" they asked. "Are you trying to introduce squishy judgmental, psychological stuff into something which we think is science?"¹²

He and his Columbia colleagues were working on totally different kinds of problems. Empowered by the Second World War, statisticians like Raiffa and Schlaifer were increasingly interested in using statistics not just to analyze data but to make decisions. In contrast, Neyman and Pearson considered the errors associated with various strategies or hypotheses and then decided whether to accept or reject them; they could not infer what to do based on an observed sample outcome without thinking about all the potential sample outcomes that could have occurred but had not. This was Jeffreys's objection to using frequentism for scientific inference. Raiffa felt the same way for different reasons; he wanted to make decisions tied to "real economic problems; not phony ones."¹³

Raiffa was interested in concrete, one-of-a-kind decisions requiring quick judgments about how much of a product to stock or how to price it. Like Schlaifer at Harvard, he wanted to help businesses resolve uncertainties and use indirectly relevant information. As Raiffa put it, anti-Bayesians "would never—and I mean never—assign probabilities to some such statement as 'the probability that p falls in the interval from .20 to .30.'"

Bayesian subjectivists, on the other hand, actually wanted answers expressed in terms of probabilities. They did not want to merely accept or reject a hypothesis. As Raiffa realized, a business owner wanted to be able to say that "on the basis of my formerly held beliefs . . . and of the specific sample outcomes, I now believe there is a .92 probability that $\tilde{p}$ is greater than .25."¹⁴

This was *verboden* for frequentists, who recognized only those sample outcomes that were "significant at the .05 level." Raiffa regarded their focus as "a very, very cursory description of the distribution. I wanted my students to think probabilistically about [the entire distribution of p , about] where

the uncertain p could be, and then figure out from the decision point of view where the correct action would be. So the whole question of a test of hypotheses seemed to me to be leading students in a wrong direction.”¹⁵

The chasm between the two schools of statistics crystallized for Raiffa when Columbia professors discussed a sociology student named James Coleman. During his oral examination Coleman seemed “confused and fuzzy . . . clearly not of Ph.D. quality.”¹⁶ But his professors were adamant that he was otherwise dazzling. Using his new Bayesian perspective, Raiffa argued that the department’s prior opinion of the candidate’s qualities was so positive that a one-hour exam should not substantially alter their views. Pass him, Raiffa urged. Coleman became such an influential sociologist that he appeared on both the cover of *Newsweek* and page one of the *New York Times*.

So far, Raiffa regarded his transformation from a Neyman-Pearsonite to a Bayesian as an intellectual conversion; his emotional conversion was still to come.

In a campaign to raise the intellectual standards of business schools, the Ford Foundation donated money to Harvard to hire a mathematical statistician in 1957. The university made Raiffa an attractive offer: joint posts in its new department of statistics and in the business school. When the statistics department learned about Raiffa’s conversion to Bayesian ideas, the chair, Frederick Mosteller, seemed tolerant but lukewarm, and another prominent professor, William Cochran, said, “Well, you’ll grow up.”¹⁷ At the business school, though, Schlaifer welcomed Raiffa with open arms.

Schlaifer was “the most opinionated person I ever met,” Raiffa recalled. At first, he did not realize “how wonderful Schlaifer was. . . . I didn’t realize that Schlaifer was as great a man as he was. He was already focused on the real business decision problems, not on testing hypotheses. He said they were getting it wrong.” Then Raiffa corrected himself: “No, no, he didn’t say that. He said they’re getting it wrong for business decision-making under uncertainty.”¹⁸

Each morning Raiffa tutored Schlaifer in calculus and linear algebra, vectors, transformations, and the like. The next morning Schlaifer would conjecture new theorems, and the day after that apply what he had learned to a concrete problem. “His mind was receptive, razor-sharp, tenacious, persistent, creative,” Raiffa discovered.¹⁹ Both men were workaholics, but Schlaifer worked longer hours than anyone else. As Raiffa recalled, “He was really a fabulous student. . . . He had raw mathematical abilities provided he could see how it might be put to use.”²⁰ The two never referred to journals or books. Everything they did together was self-generated.

Schlaifer did not know nearly as much statistics as Raiffa, but he was much better read. Raiffa had not studied the great prewar theorists Jeffreys, Fisher, and Egon Pearson. Later, when he discovered Savage's work, he was amazed at its clarity. Raiffa took the advice of his colleagues and named his Harvard chair for Frank Ramsey without ever having read the young man's work.

Writing articles with Schlaifer, Raiffa always produced the first draft. Then Schlaifer "analyzed everything seven ways to Sunday and changed the text endlessly, putting in commas, and taking them out again," recalled John Pratt, who wrote several important works with them.²¹ Schlaifer almost refused to let Harvard Business School Press publish one of his books because its editors put quotation marks outside the punctuation instead of inside. (Thus the battle raged over "over." and "over".)

"The unsettling part of him," Raiffa said, ". . . was that he made so much sense—as long as he stayed away from politics!"²² So Raiffa argued statistics with his collaborator but closed his ears when Schlaifer inveighed against the income tax and advocated solving Haiti's problems by turning all the Haitians in the United States into soldiers and sending them back home.

To Schlaifer, Bayes' rule was not just something to use, it was something to believe in—fervently. A true believer, he refused to accept that there might be several ways to approach a problem. He used "sheer brutal insistence and intellectual discourse by finding holes in every argument that Howard used," recalled his student Arthur Schleifer Jr. "He would show that these alternative ways led to untenable paradoxes. . . . Robert's approach was that there was this one way to do it, you have to do it this way, and if you do it any other way, I'll show you you're wrong."²³

Raiffa wanted to expose his students to both frequentist and Bayesian methods so that if some read the establishment's point of view they would not get confused. But Schlaifer regarded this as teaching falsehood. Anyway, he pronounced loftily, "businessmen don't read the literature."²⁴

By 1958 Raiffa had also converted passionately to subjectivism. It seemed obvious that four businesses serving four different markets could use the same information to produce four different nonstatistical priors and four different conclusions. Although this still bothers some scientists and statisticians, others were content to overwhelm their initial nonstatistical prior opinions with large amounts of new statistical information. Just as many Americans remember where they were when they heard about Kennedy's assassination or the 9/11 attack, so many Bayesians of Raiffa's generation remember the

exact moment when Bayes' overarching logic suddenly hit them like an irresistible epiphany. Critics began to call Harvard Business School "a Bayesian hothouse."

Betting—assigning different probabilities to the same phenomenon—became the tangible expression of Bayesian beliefs. "Every day there'd be a half dozen bets around [Schlaifer's group] about anything—elections and sports. Dollar bills were changing hands all the time. It was part of the ingrained way of life. You really believed this stuff," Schleifer said.²⁵ Schlaifer and Raiffa were developing reputations as zealots with a cause.

Schlaifer sent 700 pages of his first textbook to McGraw-Hill for publication as *Probability and Statistics for Business Decisions: An Introduction to Managerial Economics under Uncertainty*. Then he discovered his usual host of errors and infelicities and insisted that McGraw-Hill withdraw the first printing and replace it with a second. It was a classic case of placing intellectual rigor above business economics, and Schlaifer won. The book sold for \$11.50 in 1959, and Harvard promoted its former tutor to a professorship in business administration.

Probability and Statistics for Business Decisions was the first textbook written entirely and wholeheartedly from the Bayesian point of view. Students could solve inventory, marketing, and queuing problems using simple arithmetic, slide rules, or, at most, a desk calculator. The book acknowledged few previous authorities. Schlaifer had come to his subjectivist position independently of Ramsey, de Finetti, and Savage. In turn, Savage recognized that Schlaifer had developed his ideas "wholly independently" and was more "down to earth and less spellbound by tradition."²⁶

Mulling over the state of Bayes' rule, Schlaifer and Raiffa realized that Bayesians, unlike frequentists, had no bookshelves of mathematical tools ready for use. As a result, Bayesian methods were regarded as impractically complicated, particularly by business students, who were often mathematically unprepared. While theoreticians like Savage and Lindley tried to make Bayes mathematically respectable, Raiffa and Schlaifer set out in 1958 to make it fully operational and easy to use for bread-and-butter problems. Like George Box, they parodied a popular song, this one from Annie Get Your Gun, claiming that anything a frequentist could do, they could do better.

To make calculations easier, they introduced decision trees, tree-flipping, and conjugate priors. "I began using decision tree diagrams that depicted the sequential nature of decision problems faced by business managers," Raiffa said. "Should I, as decision maker, act now or wait to collect further market-

ing information (by sampling or further engineering)? . . . I never made any claim to being the inventor of the decision tree but . . . I became known as Mr. Decision Tree.”²⁷ Soon the diagrams of Bayes’ decision-making process were, like many-branched trees, rooted in undergraduate business curricula. The trees are probably the best-known practical application of Bayes’ rule.

Tree-flipping began as a simplification to help one of Raiffa’s graduate students who was interested in wildcat drilling for oil. Normally, a wildcatter decided whether to test a particular site *before* deciding to drill or not drill. To avoid some messy algebra, Raiffa flipped the order of the wildcatter’s decision. He dealt with the probability that test results would be positive or negative *before* he considered whether or not to conduct the test. Working through the diagram produced information about *x*’s followed by *y*’s. Tree-flipping put the *y*’s first. It amounted to using Bayes’ rule because the probability of *x* given *y* and the probability of *y* given *x* are the two critical elements of its formula.

“So you flip trees,” Raiffa said. “We didn’t call it Bayes. The worst thing you can do is to use Bayes’ theorem. It’s too complicated. Just use common sense and play around with these things, then it was pretty easy. We had people doing complicated things that could have been done by Bayes, but we didn’t do it by Bayes. We did it by tree-flipping.”²⁸

Raiffa also developed a handy shortcut for updating priors and posteriors. Called conjugate prior distributions, it used the fact that in many cases the shape or curve of a probability’s distribution is the same in both prior and posterior. Thus, if you start with normal Gaussians, you’ll end up with normal Gaussians. Conjugate priors paid dividends with the repeated updating called for by Bayes’ method. Albert Madansky used a similar concept for his H-bomb study. The shortcut would later become unnecessary with the adoption of Monte Carlo Markov Chain methods.

In a further simplification, some business Bayesians even dropped the prior odds called for by Bayes’ rule. Schleifer said, “My take on it was to forget the priors unless there was overwhelming prior evidence that you really know a lot about the parameter you’re interested in.”²⁹

Today, when TV and radio are filled with talking heads, it is hard to imagine that the use of expert opinion was *terra incognita* in the early 1960s. No one knew whether business executives would be willing to offer their opinions for incorporation into a mathematical formula. And no one was sure whether an expert’s subjective judgment would be valid. John Pratt asked his wife, Joy, whose job was promoting films in local theaters, to estimate their daily attendance. At first, her estimates fell into too narrow a range. By

comparing them with actual attendance figures—hundreds of data points taken night after night at two local theaters—Joy Pratt learned to make such accurate predictions that her husband became convinced that expert opinion could be useful. Bayesians objected that Pratt and Schlaifer analyzed the data using frequentist techniques. Bayesian methods—comparing different kinds of movies, the length of time they played, the popularity of their stars, and so on—would have been too complex. The use of expert opinion for decision making later became a major field of study.

It turned out that Joy and John Pratt were right: marketing executives risked a lot of money on the basis of very little information and loved being asked for their professional judgment. Accustomed to waiting until the end of a frequentist study to voice their opinions, they actually liked having their “managerial intuition” or “feel for a situation” folded into preliminary assessments.

Raiffa and Schlaifer began exploring such nitty-gritty questions as how to interview experts and measure their expertise. DuPont, trying to decide how big a factory to build for artificial shoe leather in 1962, was delighted to assess the prior odds of the demand for its new product. Design engineers at Ford Motor Company were equally pleased that incorporating their opinions into Bayes’ prior let Ford use smaller opinion samples. Their work opened up just about any business problem to mathematical analysis. An engineering problem might have 20 sources of uncertainty; of these, perhaps 12 could be handled by single guesses; 5 needed more testing; and 2 might be so critical that experts had to be interviewed. Bayes’ rule was solving far more complex problems than Savage’s mental exercises about curling rabbit ears.

Between 1961 and 1965 an exciting weekly seminar, generally followed by drinks in Schlaifer’s office, focused on decision making under uncertainty (DUU for short). The seminar explored utility analysis, portfolio analysis, group decision processes, theory of syndicates, behavioral anomalies, and ways to ask about uncertainties and values. Said Raiffa, “We helped shape a field.”³⁰ The seminar and two books Raiffa and Schlaifer cowrote during this period spurred the Bayesian revival of the 1960s. Raiffa was later surprised to realize that the most fertile period of his collaboration with Schlaifer had lasted only four years.

Raiffa’s and Schlaifer’s classic book for advanced statisticians, *Applied Statistical Decision Theory*, was published in 1961. Its careful, detailed analytical methods set the direction of Bayesian statistics for the next two decades. Today it sits on almost every decision analyst’s bookshelf.

When Pratt joined Raiffa and Schlaifer to write *Introduction to Statistical Decision Theory*, he soon realized that what was easy for him to do mathematically was quite difficult for Schlaifer, who could understand the mathematics but not produce it himself. By the time the book was ready for editing, Schlaifer and Raiffa had moved on to other interests. They received so many requests for their preliminary manuscript, however, that McGraw-Hill published it as a typescript in 1965. Thirty years later, Pratt and Raiffa finished it, and MIT published it as an 875-page book.

To introduce business school professors to mathematical methods, Raiffa ran an 11-month-long Ford Foundation program in 1960 and 1961. As a result, the next generation of business school deans at Harvard, Stanford, Northwestern, and elsewhere had received a heavy dose of Bayesian subjectivism for decision making, and the gospel radiated outward to schools of management. Raiffa even gave his students an 84-page handout, “An Introduction to Markov Chains,” more than 30 years before their widespread adoption by the statistical profession. By 2000 Bayesian methods were often centered in university business schools rather than statistics departments.

Raiffa and Schlaifer drifted apart after 1965. Raiffa still called himself a Bayesian who, “roughly speaking. . . . wish[es] to introduce intuitive judgments and feelings directly into the formal analysis of a decision problem.”³¹ Broadening what he knew about subjective probability, game theory, and Bayes’ rule, he left Harvard’s statistics department to take a joint chair in the business school and the economics department. There he pursued societal, rather than primarily statistical, issues in medicine, law, engineering, international relations, and public policy.

By any measure Raiffa’s move was a success. As a pioneer in decision analysis, he was one of four organizers of the Kennedy School of Government at Harvard; the founder and director of a joint East–West think tank to reduce Cold War tensions long before perestroika; a founder of Harvard Law School’s widely replicated role-playing course in negotiations; and scientific adviser to McGeorge Bundy, the national security assistant under Presidents Kennedy and Johnson. Raiffa also supervised more than 90 Ph.D. dissertations at Harvard in business and economics and wrote 11 books—no articles, only books—one of which has been in print for more than fifty years. As a Bayesian, Raiffa would cast a long shadow.

Ultimately, however, Raiffa and Schlaifer failed in their bold attempt to permeate business curricula, statistical theory, and American business life with Bayes’ rule. Schlaifer built managerial economics into a strong program

at the Harvard Business School, but Bayesian decision analysis faded from its curriculum, and Bayes' rule never supplanted "the old stuff" in American classrooms. Since the 1970s, when all the top business schools emphasized Bayesian decision theory, it has been compressed into a few weeks' study. Business students no longer do their own calculations; presumably they can hire a consultant or buy a computer program.

Many theoretical statisticians also ignored Raiffa's and Schlaifer's contributions; they were, after all, outsiders working in a business school. From his vantage point in Britain, Lindley was astonished that the statistical community paid so little attention to Schlaifer. "I was bowled over by him. The book with Raiffa is wonderful," and Schlaifer's 1971 book had computer methods "in advance of their time." Lindley considered Schlaifer "one of the most original minds that I have ever met [with] extraordinarily wide knowledge."³²

Part of their failure lay in the fact that Schlaifer remained a confirmed university theoretician. When confronted with a problem he could not solve, he set it aside and worked on something else, something business managers cannot afford to do. Nor did he consider long-term solutions and their consequences over time; he dealt in short-term results. He did little consulting work, and his lack of experience selling complicated ideas to busy executives limited the impact of Bayes' rule on working business people. He spent weeks exploring a marketing case about cottage cheese packaging in all its abstract complexities but stripped off all the textural surroundings that most caseworkers would have brought back from a field trip to a dairy. He turned his only graduate student's thesis about all the messy glory of IBM's quality control problems into a dry theoretical paper about two-stage sampling. The student's thesis ended up piled so high with abstract issues that it was not until Schlaifer went on sabbatical that Raiffa could intervene and secure the young man's Ph.D. Schlaifer was a passionate intellectual with a deep interest in narrow topics and all the time in the world for disputation.

After Raiffa moved on to other projects, Schlaifer threw himself into designing a new introductory course for Harvard's first-year students in managerial economics. Naturally, it would be based on Bayesian methods, a first in any business school. He wrote a text and titled it *Managerial Economics Reporting Control*, which he nicknamed MERC. Students hated it, called it Murk, and burned their copies on the front steps of Baker Library. When a reporter for the Harvard newspaper asked for a comment, Schlaifer replied, "Well, I'd rather be among those whose books are burned than those who burn books."

Then Schlaifer leaned intently forward: "Tell me. There is one thing

that really interests me. This book is printed on very good, very glossy paper. It must have burned very poorly. How do you burn them?”

“Well, sir,” the student answered respectfully, “we burn them page by page.”³³

Schlaifer, farseeing to the last, spent the remaining years of his life trying to write computer software for practitioners, even though teams of mathematically sophisticated programmers were already taking over the field. In 1994, at the age of 79, Schlaifer died of lung cancer. After his death, Raiffa and Pratt finished the trio’s 30-year-old opus, *Introduction to Statistical Decision Theory*. Dedicating it to their former colleague, Pratt and Raiffa hailed Schlaifer as “an original, deep, creative, indefatigable, persistent, versatile, demanding, sometimes irascible scholar, who was an inspiration to us both.”³⁴

who wrote *the federalist*?

Alfred C. Kinsey's explosive bestseller *Sexual Behavior in the Human Male* was published in 1948, the same year pollsters failed to predict Harry Truman's victory over Thomas Dewey in the presidential election. With the public crying foul, fraud, and debauchery, social scientists feared for the future of their profession. Opinion polling was one of their basic tools, so the Social Science Research Council, representing seven professional societies, appointed statistician Frederick Mosteller of Harvard University to investigate the scandals.

Mosteller's forthright report on Truman's election blamed the nation's pollsters for rejecting randomized sampling and for clinging to outdated sampling designs that underrepresented blacks, women, and the poor—all of whom voted more heavily Democratic than the population reached by the pollsters.

In the case of Kinsey's research, powerful men—including John Foster Dulles, secretary of state under Eisenhower; Arthur Sulzberger, publisher of the *New York Times*; Harold W. Dodds, president of Princeton University; and Henry P. Van Dusen, president of the liberal Union Theological Seminary—were demanding an end to funding for research on human sexuality. But Mosteller underwent Kinsey's standard interview about his sexual history and emerged impressed. Kinsey's lack of randomized sampling was statistically damning, but his work was far better than anyone else's in the field, and the country did not have 20 statisticians who could have done better. It was quietly arranged that when Kinsey wrote his next study, on the sexuality of women, Jerome Cornfield of the National Institutes of Health would help with the statistics.

Both scandals involved discrimination problems, also called classification problems, which struck at the heart of polling, science, social science, and statistics. Researchers tended to assign people or things to categories without being totally sure that the assignments were accurate or that the categories were well defined. Pollsters classified people as Republicans or Democrats; marketers divided consumers into users of one detergent or another; scientists classified plants in biology and skulls in anthropology; and social scientists categorized individuals according to personality.

Finishing up with the Kinsey committee, Mosteller looked around for a research topic involving classification issues. He had a feet-on-the-ground attitude, perhaps the result of having been raised by his divorced mother, who never graduated from high school but who had insisted, over the objections of her ex-husband, that Fred get an education. Mosteller had earned his bachelor's and master's degrees in mathematics from Carnegie Institute of Technology (now Carnegie Mellon University) in Pittsburgh and enrolled in graduate statistics at Princeton University's highly abstract mathematics department. As the primary liaison between Princeton and Columbia statisticians working on military research, he learned that he loved working on deadline on real-world problems. After the war, in 1946, Mosteller finished his Ph.D. at Princeton and, driven by his interest in health, education, and baseball, moved to Harvard. The investigations of campaign polling and sexual research left him ripe for a problem of his own choosing.

Mosteller began looking around for a large database to use for developing ways to discriminate between two cases. He began thinking—not about Bayes' rule—but about a minor historical puzzle: *The Federalist* papers. Between 1787 and 1788, three founding fathers of the United States, Alexander Hamilton, John Jay, and James Madison, anonymously wrote 85 newspaper articles to persuade New York State voters to ratify the American Constitution. Historians could attribute most of the essays, but no one agreed whether Madison or Hamilton had written 12 others.

Mosteller had learned about the problem during a summer job he held as a graduate student in 1941. Counting the number of words in each sentence of *The Federalist* papers with psychologist Frederick Williams, he discovered “an important empirical principle—people cannot count, at least not very high.” He also learned that, stylistically, Hamilton and Madison were practically twins, skilled in a complicated oratorical style popularized in 1700s England. Mosteller agreed to “leave general style as a poor bet and pay attention to words.”¹ The job was daunting because he would need a lot of

single words to supply a pool of thousands of variables. When the summer job ended and the Second World War intervened, Mosteller forgot about *The Federalist*.

After the war, he decided that *The Federalist* might fill the bill for his classification project. By 1955 he was far enough along to rope in David L. Wallace, a young statistician at the University of Chicago. In his disarmingly casual way, Mosteller asked Wallace, “Why don’t you come up and spend some time in New England this summer, and work on this little project I’ve sort of started?”² The two wound up spending more time studying *The Federalist* papers than Hamilton and Madison did writing them—“a horrible thought,” Wallace said later.

Wallace urged Mosteller to use Bayes’ rule for the project. Wallace had earned a Ph.D. in mathematics from Princeton in 1953 and would become a professor at the University of Chicago. But in 1955, his first year at Chicago, Savage was teaching from his recently published book on Bayes’ rule. Despite the hostility of most American statisticians, Wallace had been receptive to Bayesian ideas.

Wallace thought *The Federalist* might be an application where Bayes could be very helpful. “If you stick to relatively simple problems,” he explained, “like the ones taught in elementary statistics books, you can do it by Bayesian or non-Bayesian methods, and the answers are not appreciably different.” Laplace had investigated such problems in the early 1800s and discovered the same thing. “I’m not really a Bayesian,” Wallace said. “I haven’t done much more than *The Federalist*, but . . . when you have large numbers of parameters, of unknowns to deal with, the difference between Bayes and non-Bayes grows immense.”

Mosteller was open to suggestion. Unlike Savage, Lindley, Raiffa, and Schlaifer, he was no fervent Bayesian. He was an eclectic problem solver who liked any technique that worked. He accepted the validity of both kinds of probability: probability as degrees of belief and probability as relative frequency. The problem, as he saw it, was that treating a unique event like “Hamilton wrote paper No. 52” was difficult with sampling theory. Bayes’ degrees of belief would be harder to specify but more widely applicable.

In addition, Mosteller liked grappling with critical social issues, not avoiding controversy by taking refuge in textbook examples. Reality added a certain frisson to a problem. As he put it, difficulties found “in the armchair” seldom resembled those in the field or scientific laboratory. In later years, when asked why he spent so much time on *The Federalist* papers, Mosteller

would point to “that Bayesian hothouse” in the Harvard Business School and to the fact that Raiffa and Schlaifer did not deal with difficult problems or complex data. The discrepancy between too many Bayesian theories and too few practical applications disturbed Mosteller.

With Savage’s encouragement, Mosteller and Wallace started their quest to “apply a 200-year-old mathematical theorem to a 175-year-old historical problem.” In the process, Mosteller would organize the largest civilian application of Bayes’ rule since Laplace studied babies and Jeffreys analyzed earthquakes. Not coincidentally, Wallace and Mosteller would resort to so-called high-speed computers.

They had seemingly vast amounts of data: 94,000 words definitely written by Hamilton and 114,000 by Madison. Of these, they would ignore substantive words like “war,” “executive,” and “legislature” because their use varied by essay topic. They would keep “in,” “an,” “of,” “upon,” and other context-free articles, prepositions, and conjunctions. As work progressed, however, they became dissatisfied with their “rather catch-as-catch-can methods.” In a critical decision, they decided to turn their bagatelle of a historical problem into a serious empirical comparison between Bayesian and frequentist methods of data analysis. *The Federalist* papers would become a way to test Bayes’ rule as a discrimination method.

By 1960 Wallace was working full-time on developing a Bayesian analysis of *The Federalist*, working out the details for a mathematical model of their data. Given so many variables, Wallace and Mosteller would be mining the papers like low-grade ore, successively sifting, processing text in waves, and discarding words of no help. They would use numerical probabilities to express degrees of belief about propositions like “Hamilton wrote paper No. 52” and then use Bayes’ theorem to adjust these probabilities with more evidence.

Initially, they assumed 50–50 odds for the authorship of each paper. Then they used the frequencies of 30 words—one at a time—to improve their first probability estimate. In a two-stage analysis, they first looked at the 57 papers of known authorship and then used that information to analyze the 12 papers of unknown parentage. As their calculations became increasingly complicated, Wallace developed new algebraic methods for dealing with difficult integrals; his asymptotic approximations would form much of the project’s statistical meat.

Mosteller and Wallace adopted another important simplification. Instead of using the mathematical vocabulary of probabilities, they adopted

the everyday language of odds. They were expert mathematicians, but they found odds easier computationally and intuitively.

During the decade it took to analyze *The Federalist*, Mosteller kept busy on a number of fronts. His back-to-back investigations of the Truman election and the Kinsey report had turned him into the person to call when something went wrong. Over the years Harvard would ask Mosteller to chair four departments: social relations (as acting chair); statistics (as its founder); and, in the university's School of Public Health, first biostatistics and then health policy and management.

He persuaded Harvard ("VERY SLOWLY," he wrote a friend)³ to establish the statistics department. Joining the 1950s and 1960s push to apply mathematical models to social problems, he researched game theory, gambling, and learning, where Bayes' theorem itself was not used but served as a metaphor for thinking and accommodating new ideas. Eventually, Mosteller's interest in those fields waned, and he moved on to other pastures. Education remained a major interest. As part of the government's post-Sputnik push to teach students at every level about probability, Mosteller wrote two textbooks about frequentism and Bayes' rule for high school students. In 1961 he taught probability and statistics on NBC's early-morning Continental Classroom series; his lectures were viewed by more than a million people and taken for credit by 75,000. In medical research Mosteller pioneered meta-analysis and strongly advocated randomized clinical trials, fair tests of medical treatments, and data-based medicine. He was one of the first to conduct large-scale studies of placebo effects, evaluations of many medical centers, collaborations between physicians and statisticians, and the use of large, mainframe computers.

How did Mosteller juggle a massive Bayesian analysis on top of his other work? He looked tubby and rumpled, but he was a superb organizer and utterly unfazed by controversy. He was genial; he engaged critics with a touch of humor, and he seemed to believe they were entitled to opinions he disagreed with. He was also patience itself and, with a "Gee, golly, shucks smile," explained things over and over again.⁴ Doctrinaire only about grammar and punctuation, he once wrote a student about his paper, "I am in a lonely hotel room, surrounded by whiches."⁵

Mosteller was also very hard working. He once posted a sign in his office, "What have I done for statistics in the past hour?"⁶ For a short time he recorded what he did every 15 minutes in the day. He was also, as he himself pointed out, the beneficiary of a bygone era. His wife cared for everything in his life except his professional work, and several women at Harvard devoted

their careers to his success, including his longtime secretary and Cleo Youtz, his statistical assistant for more than 50 years.

Mosteller was also said to involve in his research any student who stepped within 50 feet of his office. Persi Diaconis, a professional magician for several years after he ran away from home at the age of 14, met Mosteller on his first day as a graduate student. Mosteller interviewed him in his low-key, friendly way: “I see you’re interested in number theory. I’m interested in number theory. Could you help me do this problem?”⁷ They published the result together, and Diaconis went on to have a stellar career at Stanford University. It was said that Mosteller’s last collaborator on Earth was a mountaintop hermit and that Mosteller climbed the peak and persuaded him to cowrite a book. In truth, Mosteller collaborated only with people he considered worth his time, including Diaconis, John Tukey, the future U.S. senator Daniel Patrick Moynihan, economist Milton Friedman, and statisticians like Savage. Finally, Mosteller credited his success to the fact that “somewhere along the road I got this new way of doing scholarly work in little groups of people.”⁸ Colleagues and research assistants would divide up an interesting topic, meet every week or two, pass memos back and forth, and in four or five years publish a book. Working four or five such groups at a time, Mosteller wrote or cowrote 57 books, 36 reports, and more than 360 papers, including one with each of his children.

Four years into *The Federalist* project, he and Wallace had a breakthrough. A historian, Douglass Adair, tipped them off to a study from 1916 showing that Hamilton used “while” whereas Madison wrote “whilst.” Adair’s news told them that screening for words in the anonymous 12 papers might pan out.

The problem was that “while” and “whilst” were not used often enough to identify all 12 papers, and a printer’s typographical error or an editor’s revision could have tainted the evidence. Single words here and there would not suffice. Wallace and Mosteller would have to accumulate a large number of marker words like “while” and “whilst” and determine their frequency in each and every *Federalist* paper.

Mosteller started *The Federalist* project armed with a slide rule, a Royal manual typewriter, an electric 10-key adding machine, and a 10-bank electric Monroe calculator that multiplied and divided automatically. He greatly missed a device he had enjoyed using at MIT: an overhead projector. He and Wallace soon realized they would have to use computers. Harvard had no computer facilities of its own and relied on a cooperative arrangement with MIT. Mosteller and Wallace wound up using a substantial chunk of Har-

vard's allocation. Today, a desktop computer would be faster. Also slowing them down was the fact that Fortran was only two years old, awkward, and hard to program for words. They were "straddling the introduction of the computer for linguistic counts and the old hand-counting methods, with the disadvantages of both."

Substituting student brawn for computer power, Mosteller organized an army of 100 helpers—80 Harvard students plus 20 others. For several years, his soldiers punched cards for the supposedly high-speed computer.

Programming proceeded so slowly that Youtz speeded up the search for marker words by organizing a makeshift concordance by hand. Threading electric typewriters with rolls of adding machine tape, students transcribed one word to a line, cut the tape into strips with one word per strip, sorted them alphabetically, and counted the strips. Once someone, forever unnamed, exhaled deeply and blew a cloud of statistical confetti. Within a few days, however, Youtz's typists discovered that Hamilton used "upon" twice per paper while Madison seldom used it at all.

Soon thereafter, students punching computer cards found a fourth marker word, "enough," also used by Hamilton but never by Madison. By now, Mosteller had four words pointing to Madison as the author of the disputed papers. But here again, by editing one another's papers, Madison and Hamilton could have melded their styles. As Mosteller and Wallace concluded, "We are not faced with a black-or-white situation, and we are not going to provide an absolutely conclusive settlement. . . . Strong confidence in conclusions is the most an investigation can be expected to offer." They needed to extend their evidence and measure its strength.

Venturing beyond the simplest Bayesian applications, they found themselves knee-deep "in a welter of makeshifts and approximations." Bayesian methods for data analysis were in their infancy, so they had to develop new theory, computer programs, and simple techniques, like those frequentists had developed before 1935. Discarding whatever did not work, they wound up tackling 25 difficult technical problems and publishing four meaty, parallel studies comparing Bayes, frequentism, and two simplified Bayesian approaches. The calculations became so complicated that they checked their work by hand, largely on slide rules. "Upon" was their single best discriminator by a factor of four. Other good makers were "whilst," "there," and "on." "Even a motherly eye," Mosteller wrote, could see disparities between "may" and "his."

Their biggest surprise was that prior distributions—the *bête noir* of

Bayes' rule—were not of major importance. “This was awesome,” Mosteller said. “We went in thinking that everything depended on what sort of prior information you used. Our finding says that statisticians should remove some attention from the choice of prior information and pay more attention to their choice of models for their data.”⁹

In the end, they included prior odds only because Bayes' theorem called for them, and they arbitrarily assigned equal odds to Madison and Hamilton. The prior odds turned out to be so unimportant that Mosteller and Wallace could have let their readers name them. In a timely analogy, Mosteller noted that a single measurement by an astronaut on the moon would be enough to overwhelm any expert's prior opinion about the depth of the dust on the lunar surface. So why include the prior at all? Because with controversial or scanty data, enormous amounts of observational material might be needed to decide the issue.

When Mosteller and Wallace published their report in 1964 they announced happily that “we tracked the problems of Bayesian analysis to their lair and solved the problem of the disputed *Federalist* papers.” The odds were “satisfyingly fat” that Madison wrote all 12 of them. Their weakest case was No. 55, where the odds were 240 to 1 in Madison's favor.

Their publication was the fourth about Bayes' rule to appear in a three-year period. It followed works by Jeffreys, Savage, and Raiffa and Schlaifer. Of all these works, only Mosteller and Wallace's had dared treat real issues with Bayesian statistics and modern computers. Mosteller had thought about *The Federalist* papers for 23 years and worked on them for ten. It would long remain the largest problem publicly attacked by Bayesian methods.

The work is still admired as a profound analysis of a difficult problem. Reviewers used words like “ideal,” “impressive,” “impeccable,” and “Herculean.” As late as 1990 it was considered the greatest single Bayesian case study.

Despite the razzle-dazzle, no one followed it up. No one—not even Mosteller and Wallace—tried to confirm its results by reanalyzing the material using non-Bayesian methods. Who else could organize committees of collaborators and armies of students to empower a 1960s computer to solve large and complex problems?

As for Mosteller himself, what was his reaction? Satisfaction, of course. But also the feeling that, as a friend said, “Hey, here's a good application, a new technique, let's try it, and then find others too.” A number of his books described Bayesian techniques, and to this day Diaconis regards Mosteller as a committed Bayesian who tried hard to get social scientists to accept Bayes-

ian methods. Yet Mosteller never again devoted an entire project to Bayes. While his famous study of poverty in collaboration with Senator Moynihan influenced public policy, it did not make major use of Bayes' rule. When a student won a prize for a Bayesian Ph.D. thesis, Mosteller wrote a congratulatory note: "I think Bayesian methods are about to take off. But then I've been saying that for twenty-five years."¹⁰

the cold warrior

The *Federalist* project impressed the still small world of professional statisticians, but John Tukey, a star from the world of Cold War spying, would give Bayes' rule the opportunity to demonstrate its prowess before 20 million American television viewers. But would the statistical community learn from Tukey's example that Bayes had come of age? That was the question.

Bayes' big chance at fame commenced in 1960 with the race between Senator Kennedy and Vice President Richard M. Nixon to succeed Eisenhower as president. The election was far too close to call, but the nation's three major television networks competed fiercely to be the first to declare the victor. Winning the race would translate into prestige and advertising dollars. For the National Broadcasting Corporation (NBC), there was a bonus: the opportunity to show off the latest computers made by its corporate owner, Radio Corporation of America (RCA).

NBC's Huntley–Brinkley Report, the nation's top-rated TV news program, reached 20 million viewers each weeknight. Co-anchors Chet Huntley, broadcasting from New York, and David Brinkley, from Washington, were celebrities; more people could recognize them than Cary Grant or James Stewart. NBC's fast-paced format and informal nightly sign-off—"Good night, Chet," "Good night, David"—transformed TV news.

Despite the program's popularity, memories of the polling industry's spectacularly poor performances in the 1936 and 1948 elections as well as the extraordinarily close Nixon–Kennedy race made network executives nervous. In preparation for Election Day, NBC went looking for someone to

help it predict the winner. In the first of a series of surprises, the network approached a Princeton University professor, John W. Tukey.

Today Tukey is best known for the terms “bit” and “software,” and few outside statistics and engineering recognize his name. But he was a man of staggering accomplishments in the cloak-and-dagger world of military research, especially in code breaking and high-tech weaponry. He worked two jobs 30 miles apart: at Princeton University, where he was a professor of statistics, and at AT&T’s Bell Laboratories, then widely considered the finest industrial research laboratory in the world. From these vantage points, he advised five successive U.S. presidents, the National Security Agency, and the Central Intelligence Agency.

To appreciate the audacity of NBC’s job offer to Tukey, one needs to understand how deeply he was embedded in Cold War secrets. He had done research on topology in the late 1930s and military analysis in the 1940s. As a young man during the Second World War, Tukey worked in Princeton with the Operations Research Group that computed how a B-29 bomber speeding over Europe should aim its machine-gun fire. With pencil and paper during the Cold War, he broad-brushed the aerodynamics, trajectory, and warhead for the Nike, the first anti-aircraft surface-to-air missile system. He also helped persuade Eisenhower to build the U-2, the spy plane that flew from 1956 until 1960, the year a U-2 pilot Francis Gary Powers was shot down over the USSR.

When NBC News approached Tukey in 1960, he had been a member of the CIA’s Science and Technology Advisory Panel and the National Security Agency’s Science Advisory Board for eight years. His most famous advisory role had occurred the year before, when, as a delegate to the US–USSR Conference on the Discontinuance of Nuclear Weapon Tests, he surprised the Soviet delegation by showing that seismogram data could distinguish underground nuclear explosions from earthquakes. Once both sides knew they could police each other’s compliance, they signed the Partial Test Ban Treaty in 1963 to ban nuclear tests in the atmosphere, space, and sea.

Tukey also helped establish a super-secret cryptography think tank at Princeton University. The communications research division of the Institute of Defense Analyses (IDA) moved into Von Neumann Hall, a new campus building surrounded by an eight-foot-high brick wall. IDA, which had “the most intimate ties” to the National Security Agency, was created to solve advanced cryptographic problems.¹ Although the position did not appear on his curriculum vitae, Tukey served on IDA’s board of trustees for decades. Student protests against secret research in universities forced IDA off campus

in 1970, despite Tukey's personal appeal to Princeton's president, Robert F. Goheen.

Many university faculty did classified work as part of their regular duties during the 1950s and 1960s. John Pratt and Stephen Fienberg, for example, were cleared for such work at the University of Chicago. Said Fienberg, "When I joined the faculty in the Department of Statistics in 1968, it had a dual contract with the Navy office of research. One part supported basic statistical research, and the other was for statistical consulting. We had a safe in the basement where they kept the classified consulting work, although I do not know which faculty had been working on it."²

Tukey also worked closely with members of Princeton's physics department, which was "highly involved in the design of the atomic and later hydrogen bombs."³ After the United States dropped atomic bombs on Japan in 1945, the director of the Manhattan Project, Gen. Leslie R. Groves, asked the Princeton physics chair Henry Smyth to write the official explanation of the bomb, *Atomic Energy for Military Purposes*. In 1951 Princeton launched a secret undertaking, Project Matterhorn, to design thermonuclear weapons at its nearby Forrestal Research Center. Tukey evaluated Edward Teller's and Stanislaw Ulam's design for the first H-bomb early that year. According to his curriculum vitae, Tukey served Forrestal Research Center as "supervisor, Military Systems Analyst" from 1951 until 1956.⁴ Physics professor John A. Wheeler, who headed the weapons program, stated, "I believe that the whole country—scientifically, industrially, financially—is better off because of him and bears evidence of his influence."⁵

In addition to his military research at Princeton, Tukey taught classes and supervised more than 50 graduate students. In appearance he could be "a bouncy and beefy extrovert" and "sort of cherubic looking with a pleasant manner." But his lecture style was oblique at best. Invited to speak at Imperial College in London in 1977, Tukey looked like a great bear of a man in old baggy pants. Sitting in a cross-legged Buddha pose on the podium, he began his lecture by asking slowly and deliberately, "Comments, queries, suggestions?"⁶ During the long wait that ensued, he ate prunes—12 of them, one by one—until someone in the audience finally asked if he could explain something or other. Only then did Tukey begin speaking. When a graduate student asked one January for an appointment to discuss his Ph.D. thesis, Tukey checked his diary and said he was going to a meeting in two months and if the student drove him there they could talk about it in the car.

Tukey also advised the federal government on a wide range of civilian

problems: air quality, chemical pollution, ozone layer depletion, acid rain, census methodology, and educational testing.

How did he manage all this? Stories are legion about Tukey sitting in the back row at a seminar, dozing, reading mail, scanning newspapers, or editing articles, but then rising at the end of the talk to critique it. Tukey drafted articles in pencil while listening to baroque brass recordings, topped the article with the words “By _____ and John W. Tukey,” gave the manuscript to one of his two longtime secretaries, and then searched for a collaborator to finish the piece. He put his name to about 800 publications and worked with more than 105 coauthors, including Jerome Cornfield at NIH, but most frequently with his friend Fred Mosteller at Harvard.

As a result of Tukey’s heavy military and teaching workloads, his future father-in-law fully expected him to whip out pad and pencil while waiting at the altar to be married. His bride, Elizabeth R. Rapp, was personnel director of the three-year-old Educational Testing Service. Later, she confided that “as the wife of [a] dedicated workaholic, I understand the selfless love and devotion, accommodation and deprivation required to ‘keep them on the road.’” After Elizabeth’s death in 1998, Tukey said, “One is so much less than two.”

According to Elizabeth, Tukey organized and simplified his personal life like “a New Englander through and through.”⁸ His conversation was quiet and measured and excluded personal comments and idle chatter. His nephew Frank R. Anscombe contended that Tukey had few wants, although they included a house near the sea, a convertible, a small catamaran, classical music recordings, and mince or apple pie. Tukey traveled with his personal table tennis paddle; collected some 14,000 mystery, sci-fi, and adventure paperbacks; lunched on fistfuls of cheese and six glasses of skim milk; and drove a 1936 wood-paneled station wagon until the passenger door fell off and his papers flew onto Nassau Street in Princeton. For 40 years he wore the same style of black polo shirt, so wrinkled that students sometimes mistook him for a janitor. But he always seemed able to squeeze in one more project, provided it was sufficiently intriguing.

So how, given Tukey’s eminence and his time commitments, could NBC convince him that the Huntley–Brinkley news program warranted his attention? First, the reputation of opinion surveys, the mainstay of social science, was abysmal. Although sampling forms the foundation of statistics, commercial pollsters were painfully slow to adopt probabilistic random sampling. Serving on the Kinsey Report study committee with Mosteller, Tukey said he would prefer a random sample of three to a Kinsey sample of 300; Kinsey’s

wife said she wanted to poison him. If Tukey aimed to improve statistical practices in the polling industry, NBC News was a high-profile place to start.

Second, NBC's RCA computers may have been a draw. If Tukey accepted NBC's offer, he would not need an army of students to snip pieces of adding machine paper. RCA was a major military contractor as well as a giant in communications; it manufactured highly regarded mainframe computers for the military and big business. During the 1940s the company's large research laboratory had designed and built the Selectron memory tube for early computers, including von Neumann's Johnniac.

The opportunity to use RCA's computers to analyze election data must have been tempting. Tukey had foreseen the intimate connection between computers and statistics years earlier. When von Neumann designed an electronic computer for the Institute for Advanced Studies at Princeton in late 1945, Tukey was the only Princeton University representative on the committee and helped design the computer's architecture and electronic adding circuit. Still, Tukey's "most striking relationship with the computer was that he didn't use it"; his hardware consisted of pencil and paper.⁹

Polling reform and powerful computers, however, may have paled in importance next to the allure of NBC's vast amounts of voting data. As an undergraduate at Brown University, Tukey had majored in chemistry with doses of physics and geology, and his Ph.D. from Princeton, earned in 1939, was in topology, among the purest branches of abstract mathematics. Military research during the Second World War turned him into a "data analyst" committed to fighting the "mental rigidity" and "ossifications" of pure mathematics and abstract statistics and to bridging the gap between mathematics and science.¹⁰ The war moved him far beyond the statistician's early role as a passive observer.

After the war Tukey decided he wanted to drive "the rocky road of real problems in preference to the smooth road of unreal assumptions, arbitrary criteria, and abstract results without real attachments."¹¹ To do so, he accepted joint positions a half hour apart at Princeton University and Bell Labs. Later, whenever he was offered professorships at other universities he would ask, "Where could I ever find another Bell Labs?"¹² Like Mosteller, he preferred exploring reality, and NBC News had plenty of that.

But of all the enticements NBC could offer—restoring polling's reputation, fast computers, and real data—the most important must have been the thrill of the chase. To beat other networks to the draw, he would have to work at top speed under international scrutiny to make sense of vast amounts of

incomplete, uncertain information. It would be, as he put it later, “the best education in real-time statistics that anybody could have.”¹³ So the military consultant to presidents joined NBC’s Huntley–Brinkley Report.

Tukey’s first evening on the job, November 8, 1960, started smoothly. The race between Kennedy and Nixon was the tightest since 1916, and Kennedy would win by 120,000 votes out of 70 million cast. By 2:30 a.m., though, Tukey and his colleagues were ready to call him the winner. The pressure was too much for NBC. Network executives hustled the statisticians into a room without telephones, locked them in and refused to let them out until 8 a.m. Tukey and his team twiddled their thumbs all night, unable to release their results until morning, when it was clear Kennedy had won. Still, Tukey had prevented NBC from mistakenly declaring Nixon the winner. Relieved and impressed, the network asked him to assemble a team for the congressional election in 1962. He would work for NBC News for 18 years.

Tukey’s handpicked group eventually included ENIAC coinventor John Mauchly; Cornfield of NIH; Richard F. Link, Tukey’s first graduate student and pollster Louis Harris’s chief statistician; Yale psychology professor Robert Abelson; and David Brillinger, later a professor of statistics at Berkeley. When David Wallace finished Mosteller’s analysis of *The Federalist* papers, he too joined the group.

Wallace arrived expecting a vacation from Bayes’ rule because Tukey was thought to look down on it. Tukey is not known to have ever published anything using Bayes’ rule, and in an often-quoted remark, he said, “There are many classes of problems where Bayesian analyses are reasonable, mainly classes with which I have little acquaintance.”¹⁴ Among those were business decision making, the bailiwick of Howard Raiffa and Robert Schlaifer. The lack of methodology for quantifying Bayes’ initial prior especially irritated Tukey. Publicly, he was a data analyst who was anti-Bayesian and even antiprobability.

Thus when Wallace joined Tukey’s NBC team in 1964 he was surprised to find Bayes’ rule securely ensconced in the computing program: “I immediately thought, this is all very Bayesian. Also, I did a lot of the coding for a lot of the models over the next decade and a half, and, as far as I’m concerned, I was using Bayesian things.”¹⁵ Those who agree include Brillinger, who later became Tukey’s biographer and the editor of his papers, Pratt from Harvard, and Fienberg from Carnegie Mellon. Said Fienberg, NBC polling used “a form of empirical Bayes, where the past results were used to construct the prior distribution.”¹⁶

Nevertheless, in almost two decades of election forecasting Tukey never

admitted to using Bayes' rule. Why would someone who publicly disdained Bayes' rule and seemed to look down on it use it for something as important as announcing the next president of the United States?

Many colleagues stress that, despite appearances, Tukey was "a very private man." His nephew called him an "elliptical and enigmatic Delphic Oracle in a black polo shirt." Wallace agreed: "Tukey could be close-mouthed. . . . He was a man of extreme power and brilliance and in some ways enigmatic about himself. . . . He didn't always let everyone know what his left hand was doing. He'd deny anything Bayesian in the NBC polling."¹⁷

His dominating personality could be intimidating. While George Box was giving a seminar at Princeton University, Tukey thought he knew what Box was going to say and kept chiming in with his own commentary. Box finally asked for a show of hands. Who wanted Tukey to continue interrupting, and who wanted him to stop? When Box won, Tukey looked surprised. "In some ways, he was a very clever eight-year-old," Box recalled. "He didn't seem to understand very much about interpersonal relations." Some colleagues pointed to his early years as a child prodigy home-schooled by his mother and said that his wife, Elizabeth, helped "warm him up." Edgar Gilbert from Bell Labs concluded, "He was a very likeable personality but he was hard to understand." Peter McCullagh, an Irish statistician at the University of Chicago, called him a "constructive scientific anarchist, . . . a cultural phenomenon, revered by some, feared by others, understood by few." Part of the problem, Pratt said, was that "Tukey could argue on both sides of anything, and you didn't know where he stood."¹⁸

Adding to the confusion was Tukey's acceptance of one of Savage's most controversial tenets: subjectivity. Tukey called objectivity "an heirloom" and "a fallacy. . . . Economists are not expected to give identical advice in congressional committees. Engineers are not expected to design identical bridges—or aircraft. Why should statisticians be expected to reach identical results from examinations of the same set of data?"¹⁹

If Tukey could be hard on Bayes' rule, he was even tougher on Fisher. Tukey believed that Fisher's frequency-based ideas dated from "the world of infancy . . . the childhood of experimental statistics, a childhood spent in the school of agronomy. . . . Almost invariably, when closely inspected, data are found to violate [the] standard assumptions" required by frequency. "Far better an approximate answer to the *right* question, which is often vague, than an *exact* answer to the *wrong* question, which can always be made precise." Tukey publicized how even slight deviations from the normal model could muddle

the methods of Fisher, Neyman, and Egon Pearson. He particularly scorned frequentist “techniques for assessing significance and asserting confidence. . . . By and large, the great innovations in [frequency-based] statistics have not had correspondingly great effects upon data analysis.” Hard words indeed.²⁰

So where did Tukey stand? Anti-Bayes and anti-frequentism? Friends contend that, like Mosteller, he opposed any monolithic philosophy. Brillinger thought Tukey was annoyed, “not with Bayesian arguments per se; . . . [but] with some of the Bayesians.” Tukey said, “Discarding Bayesian techniques would be a real mistake; trying to use them everywhere, however, would in my judgment, be a considerably greater mistake.” The issue was knowing when and where. He often complained about “a natural, but *dangerous* desire for a unified approach,” explaining that “the greatest danger I see from Bayesian analysis stems from the belief that everything that is important can be stuffed into a single quantitative framework.”²¹

Tukey used almost the same language about Fisher’s fiducial alternative to Bayes. While courting his wife, Tukey confided that his mission in life was to emulate Fisher by developing methods for analyzing experimental science. But after writing 64 pages in a search for the logical foundations of Fisher’s fiducial probability, Tukey decided that “the belief in a unified structure for inference is a dangerous form of hubris.” When Tukey visited Fisher at his home in England and began asking questions about his methods, Fisher stalked angrily away, leaving the Tukeys to find their way out of his house alone. In another version of the story, Fisher threw Tukey out of his office after telling the young man that his paper was a “long screed” and that he would understand probability statements only “if you could ever get your bullheaded mind to stop and think.” In both stories, an unstoppable force met an immovable object.²²

For Tukey the only thing that mattered was the data—shorn of computerization, mathematization, probability, and theory. He named his approach exploratory data analysis (EDA). Like Bayesians, many of its proponents were ridiculed and had trouble finding jobs.

So how did Tukey resolve his paradoxical use of Bayes’ rule without admitting it? He called it something else. While Brillinger and Wallace called their NBC polling Bayesian, Tukey said it was “borrowing strength.”²³

“Anything he did, he’d call something else,” Wallace said, even if it already had a straightforward, well-established name. New names drew attention to ideas, and a colleague counted 50 terms coined by Tukey. Among those that stuck are linear programming, ANOVA, and data analysis. In one article,

Mosteller had difficulty talking him out of using musical notation—sharps, flats, and naturals. Another colleague threatened to call him J. W. Cutie for terms such as “saphe cracking,” “quefreny,” and “alanysis.” As Wallace said, “It was not always the best way to win friends and influence people. . . . But when I talked to Tukey, I essentially tried to use his terminology.”

Still, Bayes’ rule by any other name is Bayes’ rule. And both Tukey and Mosteller were willing to use whatever statistical tool was needed, even if it was Bayesian. Beginning work long before Election Day, Wallace built a base of initial information by combining data from preelection polls; non-statistical, expert opinion from political scientists; and the voting histories of precincts, counties, cities, and states. Preelection opinion polls did not always ask the right questions, so they often failed to elicit all the information needed. The work of sampling people, surveying them, analyzing their answers, and summarizing the results was complex.

On election night, as partial returns from counties and complete returns from selected precincts flowed in, Tukey and his colleagues watched for swings and deviations from past voting behavior and from political scientists’ opinions. Then they modified their initial odds with the new information.

As Wallace relived the moment, he said, “Say we’re working at a county level with data coming in. Suppose you had no returns from one county. A strict non-Bayesian would say, ‘I can’t tell you anything there,’ but a slightly Bayesian person would say, ‘I don’t know what’s happening in county A, but county B is very similar and it’s showing a swing 5% toward Republicans.’ You might say that county A might be going the same way, but not give it great weight, because you do have to come up with a number. . . . ‘Okay,’ says Tukey, ‘go down low, take a group of counties that are similar, weight the data you get in these counties, give zero weight to non-data counties, and upgrade, update it all the time.’” Like Schlaifer at the Harvard Business School, Tukey had concluded that since he had to make a decision with inadequate information, whatever knowledge existed was better than nothing.

Wallace continued: “You take information where you have it and compute it with a lot of error bounds on it into places where you don’t have data. . . . You first of all work at rural area counties, then urban areas, north, south areas, or whatever, do it separately and play this game of upward regions and across states. It’s ‘borrowing strength,’ but I’d say it’s Bayesian. . . . You’re using historical data, from previous elections, to show variability between counties and that’s the source of your priors, so it’s very Bayesian, with a hierarchical model and historically based prior variances.”

Despite weeks of planning and rehearsals, election nights did not always proceed as hoped. Studio 8H in Rockefeller Center, where Huntley sat, was sacrosanct and off-limits to anyone without a special ID badge. But when Brillinger saw that the ID tag resembled the sugar packets in NBC's canteen, he paperclipped a packet to his shirt and wandered happily around. For the election of 1964 between Lyndon Johnson and Barry Goldwater, NBC showcased seven of its mainframe computers on Studio 8H's stage floor: several of RCA's early 301 models and two spanking new 3301's. All evening viewers could see their imposing large black boxes on screen behind Huntley. Unfortunately, the computers did not work that night, either because their operating system was unfinished or because the heat of the recording studio's lights fried them. So there they sat all night long, like so many limp rags, Link thought, impressive but useless. With voting results pouring in from around the country, Tukey's team punched furiously away at old-fashioned hand calculators and adding machines. Fortunately, the work was simple that night because LBJ's victory was a foregone conclusion, and Johnson won with a record 61% of the popular vote.

In another election, the team called the winners early for California and New York. Then, late-arriving figures came in contradicting their announcement. Two tense hours passed before voting patterns moved back in line with their predictions. Another extremely tight election kept them at work from Tuesday afternoon straight through to Thursday afternoon. Tukey and Wallace realized they needed to improve their technique.

"It turned out that the problem of projecting turnout was more difficult than that of projecting candidate percentage," Wallace discovered. "The quality of data you get is dubious, and you have clear biases of reporting coming in from one part of the county, and if you're lucky, they might do it randomly but machine votes come in faster than non-machine votes. Sometimes there's chicanery as well. The turnout is very hard to predict and that has a startling effect, so Bayes was not a total serving. I had a conversation with a student who was consulting with one of the other networks and he said to me, 'This is just so wonderful because it's the first place in statistics where all your assumptions are totally valid.' I was appalled. . . . It's a highly biased sample. And you're going to get yourself into serious trouble if you don't realize that."²⁴

Tukey continued to work for NBC through the election in 1980. After that, NBC switched to exit polls based on interviews of voters emerging from their precincts. Exit polls were cheaper, more photogenic, personal,

and chatty. They were the polar opposite of Tukey's secret, highly complex, mathematized approach.

Then came the biggest surprise of all. Like Churchill's muzzle on post-war Bletchley Park, Tukey refused to let any of his colleagues write or even give talks about their polling methods. He never wrote about them either. He said they were proprietary to RCA.

Why the secrecy? Why did Tukey scorn Bayes' rule in public but use it privately for two decades? Toward the end of his life he conceded that "probably the best excuse for Bayesian analysis is the need . . . to combine information from other bodies of data, the general views of experts, etc., with the information provided by the data before us."²⁵ He even defended Savage's subjectivist gospel, that people could look at the same information yet reach different conclusions. Reject "the fetish of objectivity," Tukey declared.²⁶ He also used a Bayesian argument when testifying before Congress that the U.S. Census should adjust for its undercounts of minorities in some areas by incorporating information from other, similar regions. As of 2010, the Census Bureau had not done so.

So why didn't—or couldn't—Tukey use the B-word? As Brillinger noted, "Bayes is an inflammatory word."²⁷ Certainly, Tukey's term "borrowing strength" allowed him to avoid it. Perhaps sidestepping it carved out a neutral workspace. Perhaps he felt the need to put his own stamp on another person's work. Or perhaps there was another reason. Given Tukey's personality, it's difficult to know. Halfway through his stint with NBC, RCA gave up trying to compete with IBM and sold off its computer division to Sperry Rand. After that, why would RCA care whether Tukey's system went public? Could RCA's military sponsors have classified Tukey's methods, and was he using Bayes' rule for his classified cryptographic research?

Many details of Tukey's national security career remain "murky, deliberately so on his part," his nephew Anscombe concluded.²⁸ But as Wallace says, "If you go to the secret coding agencies, you'd find that Bayes had a larger history. I'm not in a position to speak of that but I. J. Good is the principal contributor to the Bayesian group and he was taking that position."²⁹ Good was Alan Turing's cryptographic assistant during the Second World War. So did Tukey use Bayes' rule for decoding for the National Security Agency? And could he have been distancing himself from Bayesian methods in order to protect work there?

The ties between Tukey, Bayes, and top-secret decoding are many and close. Bayes' rule is a natural for decoders who have some initial guesses and

must minimize the time or cost to reach a solution; it has been widely used for decoding ever since Bletchley Park. Tukey's ties to American cryptography were particularly tight. According to William O. Baker, then head of Bell Labs, Tukey was part of the force that helped decrypt Germany's Enigma system during the Second World War and Soviet codes during the Cold War. Tukey served on NSA's Science Advisory Board, which was devoted to cryptography. It was a ten-member panel of scientists from universities, corporate research laboratories, and think tanks; they met twice yearly at Fort Meade, Maryland, to discuss the application of science and technology to code breaking, cryptography, and eavesdropping. Baker, Tukey's close friend, was probably the committee's most important member. Baker chaired a long study of America's coding and decoding resources for the NSA and called for a Manhattan Project-like effort to focus, not on publishable and freely available research, but on top-secret studies. Whether Tukey actually did hands-on cryptography is not known, but as a professional visiting-committee consultant, he was certainly aware of all the statistical methods being used.

Tukey's relationship with Good, one of the leading Bayesians and cryptographers of the 1950s and 1960s, is also suggestive. Tukey visited Good in Britain and invited him to lecture at Bell Labs in October 1955. The day after Good's talk he was surprised to find that Tukey, lying on the floor to relax, had obviously understood it all. Tukey was also sympathetic enough with Good's Bayesian methods to introduce him to Cornfield at NIH and suggest that Good might help with statistical methods there; Cornfield became a prominent Bayesian.

Claude Shannon was also in the audience during Good's talk. Shannon had used Bayes' rule at Bell Labs for his pathbreaking cryptographic and communications studies during the Second World War. Tukey was close to Shannon; in 1946 Tukey coined the word "bit" for Shannon's "binary digit." Tukey, Shannon, and John R. Pierce applied together for a patent for a cathode ray device in 1948.

The evidence is substantial enough to convince some of Tukey's colleagues, including Judith Tanur and Richard Link, that he probably did use Bayes' rule for decoding at Bell Labs. Brillinger, Tukey's biographer and NBC polling colleague, concluded, "I have no problem thinking that he might have."³⁰

Whatever the motivation, Tukey's secrecy edict played a major role in the history of Bayes' rule. As Wallace observed, "It's important to the development of Bayesian statistics that a lot was under wraps."³¹ Tukey's censorship

of his polling methods for NBC News, like the highly classified status of Bayesian cryptography during and after the Second World War, is one reason so few realized how much Bayes' rule was being used.

Tukey's Bayesian polling—conducted in the glare of international publicity for two of the most popular TV anchors of the day—could have spread the news of Bayes' power and effectiveness and reinforced it at regular intervals. But his ban on speaking or writing about it meant that Bayes' rule played a starring role on TV for almost two decades—without most statisticians knowing about it.

As a result, the only large computerized Bayesian study of a practical problem in the public domain during the Bayesian revival of the 1960s was the Mosteller–Wallace study of *The Federalist* in 1964. It would be 11 years before the next major Bayesian application appeared in public. And after Tukey stopped consulting for NBC in 1980, it would be 28 years before a presidential election poll utilized Bayesian techniques again.

When Nate Silver at FiveThirtyEight.com used hierarchical Bayes during the presidential race in November 2008, he combined information from outside areas to strengthen small samples from low-population areas and from exit polls with low response rates. He weighted the results of other pollsters according to their track records and sample size and how up to date their data were. He also combined them with historical polling data. That month Silver correctly predicted the winner in 49 states, a record unmatched by any other pollster. Had Tukey publicized the Bayesian methods used for NBC, the history of political polling and even American politics might have been different.

three mile island

After years of working together, the two old friends Fred Mosteller and John Tukey reminisced in 1967 about how “the battle of Bayes has raged for more than two centuries, sometimes violently, sometimes almost placidly, . . . a combination of doubt and vigor.” Thomas Bayes had turned his back on his own creation; a quarter century later, Laplace glorified it. During the 1800s it was both employed and undermined. Derided during the early 1900s, it was used in desperate secrecy during the Second World War and afterward employed with both astonishing vigor and condescension.¹ But by the 1970s Bayes’ rule was sliding into the doldrums.

A loss of leadership, a series of career changes, and geographical moves contributed to the gloom. Jimmie Savage, chief U.S. spokesman for Bayes as a logical and comprehensive system, died of a heart attack in 1971. After Fermi’s death, Harold Jeffreys and American physicist Edwin T. Jaynes campaigned in vain for Bayes in the physical sciences; Jaynes, who said he always checked to see what Laplace had done before tackling an applied problem, turned off many colleagues with his Bayesian fervor. Dennis Lindley was slowly building Bayesian statistics departments in the United Kingdom but quit administration in 1977 to do solo research. Jack Good moved from the super-secret coding and decoding agencies of Britain to academia at Virginia Tech. Albert Madansky, who liked any technique that worked, switched from RAND to private business and later to the University of Chicago Business School, where he claimed to find more applications than in statistics departments. George Box became interested in quality control in manufacturing and, with W. Edwards Deming and others, advised Japan’s automotive industry. Howard

Raiffa also shifted gears to negotiate public policy, while Robert Schlaifer, the nonmathematical Bayesian, tried to program computers.

When James O. Berger became a Bayesian in the 1970s, the community was still so small he could track virtually all of its activity. The first international conference on Bayes' rule was held in 1979, in Valencia, Spain, and almost every well-known Bayesian showed up—perhaps 100 in all.

Gone was the messianic dream that Bayes' rule could replace frequentism. Ecumenical pragmatists spoke of synthesizing Bayesian and non-Bayesian methods. The least controversial ideal, Mosteller and Tukey agreed, was either a frequency-based prior or a "gentle" prior based on beliefs but ready to be overwhelmed by new information.

When Box, J. Stuart Hunter, and William G. Hunter wrote *Statistics for Experimenters* in 1978, they intentionally omitted any reference to Bayes' rule: too controversial to sell. Shorn of the big bad word, the book was a bestseller. Ironically, an Oxford philosopher, Richard Swinburne, felt no such compunctions a year later: he inserted personal opinions into both the prior hunch and the supposedly objective data of Bayes' theorem to conclude that God was more than 50% likely to exist; later Swinburne would figure the probability of Jesus' resurrection at "something like 97 percent." These were calculations that neither the Reverend Thomas Bayes nor the Reverend Richard Price had cared to make, and even many nonstatisticians regarded Swinburne's lack of careful measurement as a black mark against Bayes itself.

Throughout this period Jerzy Neyman's bastion of frequentism at Berkeley remained the premiere statistical center of the United States. Stanford's large statistics department, bolstered by Charles Stein and other University of California professors who had refused to sign a McCarthy-era loyalty oath, was also enthusiastically frequentist, and anti-Bayesian signs adorned professors' office doors.

Bayesians were treading water. Almost without knowing it they were waiting until computers could catch up. In the absence of powerful and accessible computers and software, many Bayesians and anti-Bayesians alike had given up attempts at realistic applications and retreated into theoretical mathematics. Herman Chernoff, whose statistical work often grew out of Office of Naval Research problems, got so impatient with theoreticians spinning their wheels on increasingly elaborate generalizations that he moved from Stanford to MIT in 1974 and then on to Harvard. "We had reached a period," he wrote, "where we had to confront the computer much more intensively and we also had to do much more applied work . . . I thought, for the future,

the field needed a lot more contact with real applications in order to provide insights into which way we should go, rather than concentrating on further elaborations on theory.” Chernoff was no Bayesian, but he told statistician Susan Holmes, then beginning her career, how to face difficult problems: “Start out as a Bayesian thinking about it, and you’ll get the right answer. Then you can justify it whichever way you like.”²

Within Bayesian circles, opinions were still defended passionately. Attending his first Bayesian conference in 1976, Jim Berger was shocked to see half the room yelling at the other half. Everyone seemed to be good friends, but their priors were split between the personally subjective, like Savage’s, and the objective, like Jeffreys’s—with no definitive experiment to decide the issue. Good moved eclectically between the two camps.

In a frustrated circle of blame, Persi Diaconis was shocked and angry when John Pratt used frequentist methods to analyze his wife’s movie theater attendance data, because there was too much for the era’s computers to handle. But one of the low moments in Diaconis’s life occurred in a Berkeley coffee shop, where he was correcting galley proofs of an article and Lindley blamed him for using frequency methods in the article. “And you’re our leading Bayesian,” Lindley complained.³ Lindley, in turn, upset Mosteller by passing up a chance to do a big project using Bayes instead of frequency. Every opportunity lost for Bayes was a blow to the cause and a reason for recrimination. By 1978 the Neyman–Pearson frequentists held “an uneasy upper hand” over the Bayesians, while a third, smaller party of Fisherians “snipe[d] away at both sides.”⁴

Few theorems could boast such a history. Bayesians had developed a broad system of theory and methods, but the outlook for proving their effectiveness seemed bleak. De Finetti predicted a paradigm shift to Bayesian methods—in 50 years, post-2020. The frequentist Bradley Efron of Stanford estimated the probability of a Bayesian twenty-first century at a mere .15.

Politicking for Bayes in Britain, Lindley said, “The change is happening much more slowly than I expected. . . . It is a slow job. . . . I assumed in a naïve way that if I spent an hour talking to a mature statistician about the Bayesian argument, he would accept my reasoning and would change. That does not happen; people don’t work that way. . . . I think that the shift will take place through applied statisticians rather than through the theoreticians.” Asked how to encourage Bayesian theory, he answered tartly, “Attend funerals.”⁵

With Bayesian theory in limbo, its public appearances were few and

far between. Consequently, when the U.S. Congress commissioned the first comprehensive study of nuclear power plant safety, the question arose: would anyone dare mention Bayes by name, much less actually use Bayes' rule?

President Eisenhower had launched the nuclear power industry with his Atoms for Peace speech in 1953. Twenty years later, although no comprehensive study of safety risks to the public or the environment had been made, private corporations owned and operated 50 nuclear power plants in the United States. When Congress began debating whether to absolve plant owners and operators of all liability for accidents, the U.S. Atomic Energy Commission finally ordered a safety study.

Significantly, as it turned out, the man appointed to lead the study was not a statistician but a physicist and engineer. Born in Harrisburg, Pennsylvania, in 1927, Norman Carl Rasmussen had served a year in the navy after the Second World War, graduated from Gettysburg College in 1950, and earned a Ph.D. in experimental low-energy nuclear physics at MIT in 1956. He taught physics there until MIT formed one of the first departments of nuclear engineering, in 1958.

When Rasmussen was appointed to assess the safety of the nuclear power industry, there had never been a nuclear plant accident. Believing that any such accident would be catastrophic, engineers designed the plants conservatively, and the government regulated them tightly.

Lacking any data about core melts, Rasmussen decided to do as Madansky had done at RAND when studying H-bomb accidents. He and his co-authors would deal with the failure rates of pumps, valves, and other equipment. When these failure rates did not produce enough statistics either, the Rasmussen group turned to a politically incendiary source of information: expert opinion and Bayesian analysis.

Engineers had long relied on professional judgment, but frequentists considered it subjective and not reproducible and banned its use. Furthermore, the Vietnam War had ended America's enchantment with expert oracles and think tanks. Confidence in leaders plummeted, and a "radical presumption of institutional failure" took its place. Faith in technology dropped too; in 1971 Congress canceled its participation in the supersonic passenger plane, the SST, one of the few times the United States has rejected a major new technology. "No Nukes" activists were demonstrating across the country.

Lacking direct evidence of nuclear plant accidents, Rasmussen's team felt it had no choice but to solicit expert opinion. But how could they combine that with equipment failure rates? Normally, Bayes' theorem provided the

way. But Rasmussen's panel already had enough controversy on its hands dealing with nuclear power. The last thing they needed was an argument over methods.

To avoid using Bayes' equation, they employed Raiffa's decision trees. Raiffa was a Bayesian missionary, and his trees had Bayesian roots, but that did not matter. Panel members avoided even the words "Bayes' rule"; they called it a subjectivistic approach. They thought that keeping clear of Bayes' theorem would absolve them of being Bayesians.

The committee's final report, issued in 1974, was loaded with Bayesian uncertainties and probability distributions about equipment failure rates and human mistakes. Frequentists did not assign probability distributions to unknowns. The only reference to Bayes' rule, however, was tucked into an inconspicuous little corner of appendix III: "Treating data as random variables is sometimes associated with the Bayesian approach . . . the Bayesian interpretation can also be used."⁶

But avoiding the use of the word "Bayes" did not acquit the report of blame. Although several later studies approved of its use of "subjective probabilities," some of the report's statistics were roundly damned. Five years later, in January 1979, the U.S. Nuclear Regulatory Commission withdrew its support for the study. The Rasmussen Report seemed doomed to oblivion.

Doomed, that is, until two months later when the core of the Three Mile Island-2 nuclear generating unit was damaged in a severe accident. At almost the same time, Jane Fonda debuted a blockbuster movie, *The China Syndrome*, about the coverup of an accident at a nuclear power plant. The civilian nuclear power industry collapsed in one of the most remarkable reversals in American capitalism. Although approximately 20% of U.S. electric power came from 104 nuclear power plants in 2003, at this writing no new facility has been ordered since 1978.

Three Mile Island revived the Rasmussen Report and its use of subjectivistic analysis. After the accident the committee's insights seemed prescient. Previous experts had thought that the odds of severe core damage were extremely low and that the effects would be catastrophic. The Rasmussen Report had concluded the reverse: the probability of core damage was higher than expected, but the consequences would not always be catastrophic. The report had also identified two significant problems that played roles in the Three Mile Island accident: human error and the release of radioactivity outside the building. The study had even identified the sequence of events that ultimately caused the accident.

Not until 1981 did two industry-supported studies finally employ Bayes' theorem—and admit it. Analysts used it to combine the probabilities of equipment failures with specific information from two particular power plants: Zion Nuclear Power Station north of Chicago and Indian Point reactor on the Hudson River, 24 miles north of New York City. Since then, quantitative risk analysis methods and probabilistic safety studies have used both frequentist and Bayesian methods to analyze safety in the chemical industry, nuclear power plants, hazardous waste repositories, the release of radioactive material from nuclear power plants, the contamination of Mars by terrestrial microorganisms, the destruction of bridges, and exploration for mineral deposits. To industry's relief, risk analysis is also now identifying so-called unuseful safety regulations that can presumably be abandoned.

Subjective judgment still bothers many physical scientists and engineers who dislike mixing objective and subjective information in science. Avoiding the word "Bayes," however, is no longer necessary—or an option.

the navy searches

Surprisingly, given Bayes' success in fighting U-boats during the Second World War, the U.S. Navy embraced the method slowly and grudgingly during the Cold War. High-ranking officers turned to Bayes almost accidentally, hoping at first to garner only the trappings of statistics. Later, the navy would move with increasing confidence and growing computer power to fine-tune the method for antisubmarine warfare. Meanwhile, the Coast Guard eyed the method for rescuing people lost at sea. As was often the case with Bayes' rule, a series of spectacular emergencies forced the issue.

The navy's flirtation with the approach began at dusk on January 16, 1966, when a B-52 jet armed with four hydrogen bombs took off from Seymour Air Force Base near Raleigh, North Carolina. Each bomb was about ten feet long and as fat as a garbage can and had the destructive power of roughly one million tons of TNT. The jet's captain, known for smoking a corncob pipe in the cockpit, and his six-man crew were scheduled to fly continuously for 24 hours and refuel several times in midair.

In a controversial program called Operation Chrome Dome, SAC under Gen. Curtis LeMay kept jets equipped with nuclear weapons flying at all times to protect against Soviet attack. In a costly and hazardous process, tankers refueled the jets in midair.

The jet made the scheduled rendezvous for its third refueling with a SAC KC-135 tanker jet on the morning of January 17. Bomber and tanker maneuvered in tandem over Spain's southeastern coast, six miles above the isolated hamlet of Palomares, Spanish for Place of the Doves. They used a telescoping boom that required the two planes to fly three or four meters

apart at 600 miles per hour for up to half an hour. In a split-second miscalculation, the tanker's fuel nozzle struck the metal spine of the bomber, and at 10:22 a.m. local time 40,000 gallons of fuel burst into flame. Seven of the planes' 11 crew members perished.

Airmen, the four bombs, and 250 tons of aircraft debris rained down from the sky. Fortunately, it was a holiday, and most of the area's 1,500 residents were taking time off from working their fields, so no one was hit. Even more important, no nuclear explosion occurred; the bombs had not been "cocked," or activated. However, the parachutes on two of them failed to open, and when the bombs hit the ground their conventional explosives detonated, contaminating the area with an aerosol of radioactive plutonium. Three of the bombs were located within 24 hours, but the fourth was nowhere to be found.

Adding to the crisis was the fact that, unknown to the public, the incident at Palomares was at least the twenty-ninth serious accident involving the air force and nuclear weapons. Ten nuclear weapons involved in eight of these accidents had been jettisoned and abandoned at sea or in swamps, where they presumably remain to this day. The missing weapons, none of which involved a nuclear detonation, included two lost over water in 1950; two nuclear capsules in carrying cases in a plane that disappeared over the Mediterranean in 1956; two jettisoned into the Atlantic Ocean off New Jersey near Atlantic City in 1957; one left at the mouth of the Savannah River off Tybee Beach in Georgia in 1958; the one that fell in Walter Gregg's garden near Florence, South Carolina, in 1958; one in Puget Sound in Washington State in 1959; uranium buried in Goldsboro, North Carolina, in 1961; and a bomb from a plane that rolled off an aircraft carrier into the Pacific in 1965. It was an unenviable record that was only slowly attracting media attention.

When it became obvious that the latest H-bomb to fall from a SAC jet must have landed in the Mediterranean Sea, the Defense Department phoned John Piña Craven, the civilian chief scientist in the U.S. Navy's Special Projects Office.

Craven had a bachelor's degree from Cornell University's naval science training program and a master's in physics from Caltech. While working on a Ph.D. in applied physics at the University of Iowa he spent his spare time taking advanced courses of every kind, from journalism and philosophy of science to partial differential equations. Notably, in view of what was to come, he took statistics and got a C. In 1951 Craven graduated "sort of educated in everything."¹ These were the years when the military was developing crash

programs for using navigational satellites and for building ballistic missiles and guidance systems to counter the Soviets. In such an atmosphere, the Pentagon regarded any Caltech grad as a technological whiz kid.

At 31, Craven became what he called the navy's "Oracle at Delphi, . . . an applied physicist advising the Navy whenever they have mission or equipment problems that they cannot handle." His first job was inventing technology to locate Soviet mines blocking Wonson Harbor during the Korean War. Three years later he became chief scientist of the Special Projects Office developing the Polaris Fleet Ballistic Missile Submarine System. When the nuclear submarine *USS Thresher* burst and sank off Cape Cod in 1963 with 129 men on board, he was ordered to develop ways to find objects lost or sunk in deep water. To the military looking for an H-bomb in the Mediterranean Sea, Craven sounded like the man for the job.

"We've just lost a hydrogen bomb," W. M. "Jack" Howard, the assistant secretary of defense for atomic energy, said when he telephoned Craven.

"Oh, we've just lost a hydrogen bomb," Craven recalls saying. "That's your problem, not mine."

The deputy secretary persisted: "But one of the bombs fell in the ocean, so we don't know how to find it; three others are on land."

Craven shot back, "You called the Navy all right but you called the wrong guy. The Supervisor of Salvage is the guy responsible for that." Within hours, though, Craven and the salvage chief, Capt. William F. Searle Jr., formed a joint committee of rejects: Craven had failed twice to get into the naval academy, and Searle graduated from Annapolis before poor vision shunted him into underwater salvage work, where, wits said, everyone is more or less blind.

"Craven, I want a search doctrine," Searle barked. He needed the doctrine—naval-ese for a plan—so he could start work the next morning to send ships and other materiel to Spain. That night Craven kept telling himself, "Jesus, I've got to come up with a search doctrine."

Craven already knew something about Bayesian principles. His mine-sweeping mentor during the Korean War in 1950–52 was the navy physicist and applied mathematician Rufus K. Reber, who had translated Bernard Koopman's Bayesian antisubmarine studies into practical but classified tables for sea captains planning mine-searching sweeps. Craven had also learned about Bayes while visiting MIT professors doing classified research for the government. Most important, he had heard about Howard Raiffa, who was pioneering the use of subjective probabilistic analyses for business deci-

sion making, operational analysis, and game theory at the Harvard Business School.²

As Craven understood it, Raiffa used Bayesian probability to discover that horse race bettors accurately predict the odds on horses winning first, second, and third place. For Craven, the key to Raiffa's racetrack culture was its reliance on combining the opinions of people "who really know what's going on and who can't verbalize it but can have hunches and make bets on them." Later, Raiffa commented that he was pleased if he had influenced Craven to assess subjective probabilities and pool them across experts. But he emphasized that Bayes does not get into the act until those subjective views are updated with new information. Furthermore, he remembered talking about weather prediction, not horse races.

"I'm very good at grasping concepts," Craven explained later. "I'm lousy on detail. I got the betting on probabilities and also got the connection with Bayes's conditional probabilities. But I also understand the politics of getting things done in the Navy, and I say I've got to get a search doctrine."

Craven had experts galore at his disposal. Some knew about B-52s while others were familiar with the characteristics of H-bombs; bomb storage on planes; bombs dropping from planes; whether the bomb would stay with the plane wreckage; the probability that one or both of a bomb's two parachutes deploys; wind currents and velocity; whether the bomb will be buried in sand; how big it would look wrapped in its chute, and so on. Craven figured that his experts could work out hypotheses as to where the bomb would fall and then determine the probability of each hypothesis.

Most academic statisticians would have thrown in the towel. They would have believed, with Fisher and Neyman, that sources of information should be confined to verifiable sample data. Craven, of course, had no wish to repeat the experiment. He needed to find the bomb. "At that point, I wasn't looking at the mathematics, I was just remembering what I got from Raiffa."

Then reality intervened. With only a few hours and the assistance of one technician, Craven was forced to be "the guy who interviewed each one of these experts to make the bets. I'm the guy who decides who the bettors are, and I'm also the guy—let's be honest about it—who imagines what I'd say if I were the guy I can't get in touch with. So I'm doing a lot of imagineering. . . . I didn't have time to call together these people." Craven's use of expert guessing would be spectacularly subjective.

Blending hurried phone calls to experts, reports of on-site witnesses, and

his own “imagineering,” Craven came up with seven hypotheses he called scenarios:

1. The missing H-bomb had remained in the bomber and would be found in the bomber’s debris.
2. The bomb would be found in bomb debris along the path of the collision.
3. The bomb had fallen free and was not in the plane’s debris.
4. One of the bomb’s two parachutes deployed and carried it out to sea.
5. Both of the bomb’s parachutes deployed and carried it farther out to sea.
6. None of above.
7. A Spanish fisherman had seen the bomb enter the water. (This hypothesis came later, after naval commanders talked with one Francisco Simo Orts.)

Ideally, at this point, Craven would have gotten “all these scenarios and all these cats [his experts] in a room and have them make bets on them.” But with only one night before the search doctrine was needed, Craven realized, “I’m going to invent the scenarios myself and guess what an expert on that scenario would bet.”

The emergency forced Craven to cut through years of theoretical doubts about building a Bayesian prior and estimating the probability of its success: “As I did this, I knew immediately that I wouldn’t be able to sell this concept to any significant operator in the field. So I thought what the hell am I going to do? I’m going to tell them that this is based on Bayes’s subjective probability. And second, I’m going to hire a bunch of mathematicians, tell them I want you to put the cloak of authenticity on using Bayes’s theorem. . . . So I hired Daniel H. Wagner, Associates to do this.”

Daniel H. Wagner was a mathematician so absent-minded that his car once ran out of gas three times in one day. He had earned a Ph.D. in pure mathematics—none of it applied—from Brown University in 1957. Several years of working for defense contractors convinced him that rigorous mathematics could be applied to antisubmarine warfare and to search and detection work. The fact that both involved innumerable uncertainties made Bayes’ rule appealing. As Wagner put it, “Bayes’ rule is sensitive to information of all kinds . . . but *every clue has some error attached* because, were there no error, there would be no search problem: You would just go to the target and find

it immediately. The problem is that . . . you will rarely be given the value of the expected error and so you will have to deduce the location error from other information.”³

Operations research was new, but Wagner came recommended by two authorities: Capt. Frank A. Andrews (ret.), the officer who had commanded the *Thresher* search, and Koopman, by then an influential division head of the Institute for Defense Analyses, the campus-based organization for academics doing secret military research.

Going to Craven’s office to learn more about the missing H-bomb, Wagner took along the youngest and greenest of his three-man staff, Henry R. (“Tony”) Richardson, who had earned a Ph.D. in probability theory from Brown all of seven months earlier. He would be Bayes’ point man at Palomares.

As Wagner reconstructed the scene, Craven showed the mathematicians an interesting chart of the waters off Palomares. The seabed had been divided into discrete rectangular cells, and after interrogating air force experts Craven postulated the first six of his seven scenarios. Then he drew on statistical theory to weight each scenario as to its relative likelihood. His ideas were not quantitative; he had drawn a contour map with mountains of high probabilities and valleys of unlikely regions. Nor was he forthcoming about the reasons for each hypothesis. Richardson realized that, as far as Craven was concerned, he and Wagner were just number crunchers.

For Richardson, the fascinating feature of Craven’s probability map was that it was all based on initial information before any searching began. Craven had constructed a rule-of-thumb prior, the first component of Bayes’ rule. Richardson was familiar with Koopman’s search theory, but Craven’s multiple-scenario priors and the promise of Bayesian updating looked intriguing. By assuming that their probabilities would fall into bell shapes, Craven made it possible to use slide rules and desktop electromechanical calculators to develop a map of the bomb’s possible locations based on the prior information available to him. Like Laplace, he assigned different probability weights to each scenario.

Wagner and Richardson went to work in the company’s headquarters in Paoli, Pennsylvania, verifying and refining Craven’s rough calculations. A coworker, Ed P. Loane, constructed a more precise probability distribution for the H-bomb’s location by punching data into paper tape and feeding it over public telephone lines into an electronic computer in the nearby office of the Burroughs Corporation. Turning typewriter characters into graphical

displays on a teletype machine was challenging. A probability map might wind up looking like this:

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#\$\$\$

where # meant a probability between 0 and .05; \$ was a probability between .06 and .10, and so forth. Loane worked for Wagner, Associates full time while he was a part-time graduate student in applied mathematics at the University of Pennsylvania, and he wanted desperately to go to Palomares in Richardson’s place. Meanwhile, Craven gathered data from the Pentagon for Richardson to take to Spain. The young man was amazed to see Craven and other senior officers milling around, opening doors for him.

Almost daily planning sessions with the military soon convinced the mathematicians that their goal—using Bayes’ rule and updating to find the H-bomb—was not the reason they were hired. Bayes was window dressing. If the H-bomb was not found, the navy wanted to be able to prove statistically that it was not there. “The general thrust seemed to be to come up with a credible certification to the President that the H-bomb could not be found, rather than proceeding with an expectation that it could be found. Indeed, the former purpose,” Wagner concluded, “is the main reason we were brought into the act.”⁴

Richardson agreed: “My recollection of my marching orders was to statistically document the search that was being carried out and, in the event that the bomb was not found, to be able to certify to the President and Congress that everything possible was done and that it was done in a scientifically accurate and careful way. So that was pretty much what I was sent out to do. Having read Koopman’s work, and knowing that there was such a thing as optimal search based on Bayesian ideas, I was hoping to do more.”⁵

Richardson was not interested in using Bayes as a mathematical excuse for a failed expedition. He wanted to find the bomb. He flew to Spain with Captain Andrews, who had a physics Ph.D. from Yale and, after the *Thresher* search, had retired from the navy to join Catholic University’s faculty. Andrews knew the Pentagon had big doubts that the navy search team would ever find the bomb. In addition, he had been warned that, if the bomb was not located, the entire world would know that the search team “had failed professionally.” In short, the navy was on the hot seat, and careers were on

the line. "The implication was, of course, that if we did not find the weapon, nobody else could," Andrews recalled later.⁶

During the flight, Richardson briefed Andrews on Bayesian search theory. Andrews exclaimed, "Oh, if we'd only had that during the *Thresher* search."⁷ Once a large search area was divided into small cells, Bayes' rule said that the failure to find something in one cell enhances the probability of finding it in the others. Bayes described in mathematical terms an everyday hunt for a missing sock: an exhaustive but fruitless search of the bedroom and a cursory look in the bath would suggest the sock is more likely to be found in the laundry. Thus, Bayes could provide useful information even if the search was unsuccessful.

Arriving in Palomares, the men found an impoverished village so small it had no telephone and did not appear on maps or in Spain's census. Beginning in 3500 BC, mining and smelting for lead and silver had pockmarked the desert area with open shafts and, cursed with less than eight inches of rain annually and a saline water supply, agriculture was limited to winter tomatoes grown for export. The B-52's aerial explosion and wind had dusted 558 acres of the town and its fields with radioactive plutonium.

In addition to these problems, the village was under siege from a military camp of 750 Americans, complete with field laundries, bakeries, and a movie theater; an offshore fleet of up to 18 ships at a time; a Soviet trawler snooping in international waters; and scores of international reporters incensed by a blackout on news. Applying Bayes' rule would not be a textbook exercise in abstractions; it would be a high-wire operation conducted under intense scrutiny.

For four days, the U.S. and Spanish governments refused to admit that the bomber might have carried nuclear weapons of any kind. News of nuclear bombs and radioactivity leaked out only after an American sergeant shouted to the first reporter on the scene, "Hey, buddy, can you speak Spanish?"

"Sure."

"Well, tell that peasant over there to get out of that field, for God's sake. I can't make him understand a damned thing. There's radioactivity there and we've got to keep the people cleared out."

A public relations catastrophe was in the making. This was the first accident involving the widespread dispersal of radioactive material and the first that attracted widespread, highly critical scrutiny from the world's media. Within three days, reporters in Palomares knew that a nuclear bomb was missing, but six weeks passed before the U.S. Department of Defense would

confirm it. Censorship by Spain's dictator, Francisco Franco, kept news of the radioactivity off local radio stations, while broadcasts from Communist Eastern Europe spread the word. Radio Moscow announced that "the bomb is still in the sea, irradiating the water and the fish," and the Soviet government complained that the United States had broken the nuclear test-ban treaty of 1963. The on-site press corps fumed at being scooped by Radio Moscow, Pentagon-based reporters, and even *Stars and Stripes*.

Locals were understandably terrified. Tourism and exports of Spanish fruit and tomatoes collapsed. Demonstrators in Mexico City, Frankfurt, and the Philippines updated a popular song from *My Fair Lady*, "The bomb in Spain lies mainly in the drain."⁸ Adding to the pressure, the Vietnam War was escalating and U.S. military bases around the world were at stake. President Johnson phoned the Department of Defense every day demanding news on the search.

Arriving in this hotbed of tension, Captain Andrews immediately introduced Richardson to RAdm. William S. Guest, commander of the navy task force looking for the bomb. Guest was celebrated as the first U.S. aircraft carrier pilot to sink an enemy ship during the Second World War. He was notoriously stubborn and, behind his back, people called him Bull Dog. Guest understood airplanes and budgets, but not Bayes. However, he did understand Washington's message: "You will listen to Dr. Richardson and we will listen to him, so basically . . . for that reason you should pay attention yourself." Expecting an august authority, Guest had assigned the mathematician a captain's stateroom and steward. When he met Richardson, who looked even younger than his 26 years, Bull Dog harrumphed, "I didn't think we were getting a teenager."

The first thing Guest told Richardson was tongue-in-cheek—but not really. The mathematician was to prove that the missing bomb was on land because Guest's job was to look in the sea, and if it was on land, finding it would be someone else's job. Right off, Richardson declared, "I don't think I have the ability to do that."

Guest commanded 125 swimmers and scuba divers eyeballing the shallow coastline, minesweepers cruising deeper water in heavy surf, 3,000 navy personnel, 25 navy ships, 4 research submersibles, and a host of civilian researchers and contractors. The entire search, called Aircraft Salvops Med, would cost \$12 million in 1966 dollars.

Guest wanted to save money by using the equipment where it was best suited and then returning it as soon as possible. This meant he wanted to search some areas that were actually unlikely sites for the H-bomb.

Craven's initial hypotheses were based on prevailing winds, so the early hunt focused on a large rectangular area called Alpha II off the beach of Palomares. Guest ordered his swimmers, divers, and minesweepers to search there over and over again.

Richardson began work immediately by combing the charts of the search to date. The first weakness he saw was that, although there were tracks of where the ships had gone back and forth, there was no mention of effectiveness. "Just going back and forth wouldn't do you much good if you couldn't see the bottom of the ocean," he said. "And that in fact was the case. Some of their sensors couldn't penetrate deep water, so they were basically out there running around but not contributing anything to the effectiveness of the search. . . . None of this is criticism. It was just a horrible situation to be in. With the whole world looking at you, you can't tie boats up in a dock and say they're useless." So, inspired by a conversation with Andrews, Richardson coined the term "Search Effectiveness Probability" (SEP).

Looking at the map of the ocean bottom broken into a grid of little squares, Richardson computed for each of the squares the probability that, if the bomb were there, it would have been found by the amount of search effort applied to that area. "If the Search Effectiveness Probability came in at 95%, you could say to the Admiral, 'This area has been searched pretty thoroughly, and maybe you want to go somewhere else,'" Richardson said.

By then he probably knew as much as anyone about the ongoing search. Quarantined from the curious reporters on land, he worked nights in the ship's accounting office. His luggage, filled with reference books and Reber's declassified tables for minesweepers, had been lost in Madrid, so he painstakingly recreated some of the tables, overlaying bits of paper to superimpose curves. He had no alternative. Portable computers did not exist, and even IBM mainframes had only 32 kilobytes (not gigabytes or even megabytes) of memory. Armed with his paper cutouts, his slide rule, and an adding machine that could also multiply, he computed the effectiveness of each day's operations. Each morning he greeted Bull Dog Guest with new probabilities. The admiral enjoyed joking about Richardson's boyish appearance, but the probabilities unsettled him.

"I began computing SEPs—the probability you'd have found the bomb if it were there—and a lot of zeros showed up indicating that, even if it had been there, you probably wouldn't have seen it because your capabilities weren't up to the task." At the other end of the SEP scale, a "one" would have signified that the bomb would have been found, had it been there.

Richardson was calculating very few ones: “All those zeroes. When Guest saw them—remember this is some young teenager talking to him—the minute he saw zeroes, he was quite outspoken in his questions. ‘Why are you giving me zeroes when we’ve been out there for two weeks?’”

Guest began using the search effectiveness evaluations as quantitative guides for moving equipment. He wanted to document that his equipment had conducted their investigations thoroughly; he was not interested in using Bayesian updating to find new, more probable places to inspect. Even when a more likely site for the H-bomb appeared, Admiral Guest stuck to his “plan of squares.”

Years later, Craven complained that “the least informed and knowledgeable was Admiral Guest, the on-scene commanding officer.” The admiral was furious “because he thinks we’re out of our noggin.” Richardson is more forgiving. Guest “had other concerns. Bayes was a little bit high-falutin’. SEP was understandable. But you start getting into Bayesian updating and these funny words like priors and posteriors, admirals tend not to be patient with this stuff.” As a result, evaluating the search’s effectiveness became the focus of the H-bomb search. The idea of using effectiveness data to update the first Bayesian component—Craven’s presearch scenarios—faded into the background.

Meanwhile, the eyewitness testimony of the veteran fisherman Francisco Simo Orts was gaining credibility fast. The morning of the crash Orts had watched a large parachute pass over his boat and splash down 100 yards away. He called it “half a man, with the insides trailing.” Despite the odd description, his report sounded authentic. Strangely stiff in air, the object sank fast, within 30 seconds, parachute and all. Moreover, Orts said the chute was grayish; air force chutes were orange and white for personnel but gray-white for bombs. Navy personnel had interviewed Orts shortly after the crash but had discounted him because he did not use standard procedures to triangulate the spot. Having fished those waters all his life, he could make a seaman’s eye calculation of familiar mountains and villages along the shore and identify the location.

Lt. Cdr. J. Brad Mooney, assistant operations officer for deep submersibles, thought Orts might know what he was talking about. Mooney, later promoted to commander and chief of naval research, came from New Hampshire, where lobstermen used similar methods to find their submerged pots. He and Jon Lindberg, a commercial diving consultant, commandeered a jeep, found Orts in a bar, and took him to sea. When Orts twice pointed minesweepers to the same spot in the Mediterranean, Mooney believed him.

Soon Orts's testimony formed the basis for a high-likelihood hypothesis: with one parachute deployed, the bomb had plunged into a steep, deep-water canyon filled with tailings from an old lead mine. Mooney drew a one-mile radius around Orts's spot and named it Alpha I.

As Craven recalled, "We didn't find the bomb for a long period of time because the place of highest probability is a place we can't get to. It's in a narrow crevasse, too deep." Much of the military's equipment needed for a deep-sea search was inadequate: navigational charts dated from the early 1900s; detectors were "grossly inaccurate" with errors up to 1,000 yards; and many of the most useful devices were available only from commercial or research sources. They included three small submarines: the mini Alvin from Woods Hole Oceanographic Institution, the *Aluminaut* from the Reynolds Aluminum Company, and a little yellow sub called the Perry Cub.

Of Guest's entire squadron, only the two- or three-man sub Alvin could penetrate the rugged depths of the highest probability site. But Alvin's battery was fading, and to rejuvenate its power the submersible had to be lifted out of the water and docked for long periods.

Six weeks after the plane crash, Captain Andrews hitched a ride down the crevasse with the Alvin crew. Peering through its five-inch portholes, they suddenly spied a strange track leading down a slope, "totally different from anything there," Andrews recalled, "basically like somebody dragging a heavy log or barrel down the slope." Alvin's battery was running low again, so they had to abandon the skid marks and surface. Then, for two weeks a large storm circled in, grounded, and thwarted the Alvin.

During all this time President Johnson was telephoning the Department of Defense every day, only to be told, "We cannot tell you when we'll recover the bomb. We can only tell you the *probability* of when we'll recover it." Fuming, LBJ replied that he did not want a probability; he wanted a date. Privately, Craven added, "I'm sure his response was profane."

Finally, Johnson's volcanic temper exploded: "I want you to get a series of top-level academics to look at this search plan and tell me what's wrong with it. I don't want this probability stuff. I want a plan that tells me exactly when we're going to find this bomb."

Craven convened a committee of Cornell, Harvard, and MIT professors to come to the Pentagon on the morning of March 15, 1966. Number crunchers from Wagner, Associates presented "a mathematical model whose complexity defied understanding by mere mortals."⁹ The professors endorsed the Bayesian plan and adjourned for lunch.

On their return they learned that Alvin's crew, during its nineteenth dive off Palomares, had just spotted the bomb with its enormous parachute strewn over the seafloor rocks. Alvin had telephoned the surface that the H-bomb looked "like a ghost down there . . . like a big body in a shroud."¹⁰ It had hit ground in 1,300 feet of water and been dragged by currents down a steep slope almost 2,850 feet deep. It lay within a mile of where Orts had pointed.

After the bomb was safely retrieved, the fisherman sued for \$5 million in salvage prize money. At the government's request, Richardson again used optimal search theory based on Bayes' rule to estimate the value of Orts's testimony: he had saved the government at least a year's hard work. In 1971 an admiralty court in New York awarded Orts \$10,000. The United States had already settled \$600,000 on Palomares residents and given the town a \$200,000 desalting plant.

Just as the RAND Corporation's Bayesian study had warned eight years earlier, SAC's crash over Palomares diminished the authority of the U.S. Air Force. Military flights over Spain were forbidden, the number of SAC air-alert missions was halved, and responsibility for American air bases in Spain was transferred from SAC to the U.S. Tactical Air Command in Germany. In return for allowing the United States to keep its bases, Franco demanded American help in getting Spain into NATO and the Common Market.

SAC's next accident involving nuclear weapons, two years after Palomares, was the last straw for Operation Chrome Dome. The accident occurred when a B-52 loaded with four nuclear bombs crashed onto sea ice outside a U.S. air base at Thule, Greenland. The weapons were destroyed by fire, but, as at Palomares, radioactivity contaminated the area. As a result of the two accidents, the rising cost of keeping SAC's planes in the air, and the advent of intercontinental ballistic missiles, Secretary of Defense Robert McNamara ended SAC's airborne alert program in 1968.

In 2002, almost four decades after the Palomares accident, Spanish authorities said they had found no danger in the area from surface radiation. Spanish and U.S. health officials reported that no radiation-related cancers had been detected in Palomares's residents. They also said the 1,600 air force personnel who shipped 1,000 cubic meters of Palomares soil in 4,810 metal drums for burial in South Carolina had been exposed to insignificant amounts of radiation, 1/10 the current limit for radiation workers. Even though the public perceives plutonium as being extremely hazardous, government studies showed that its alpha rays are so weak they do not penetrate skin or clothing and, if ingested, pass out of the body in feces. The greatest danger

posed by plutonium occurs when it is inhaled. Despite more than 30 years of living and working in a plutonium-contaminated environment, official reports say, the residents of Palomares have inhaled far less than the maximum safe dose identified by the International Committee on Radiological Protection.

Radioactive snails discovered in 2006, however, prompted fears of dangerous levels of plutonium below ground. A joint Spanish–U.S. study was announced, and children were warned not to play in fields near the explosion sites or to eat the snails, a local delicacy.

But what about Bayes? What did it contribute to the H-bomb search? Richardson concluded that “the numbers I computed were coverage numbers so that [Guest] could say we’d covered these areas. . . . Scientifically, the big thing in my mind was that Bayes was a sidelight to the H-bomb search.”¹¹

The H-bomb hunt could have been a full-blown Bayesian exercise. The prior probabilities of Craven’s prehunt scenarios could have been updated with Richardson’s shipboard data to guide the search. However, they were never combined in time to be of any use in locating the lost bomb. And without updating, there was no Bayes. Instead of Bayes, the heroes were Orts and the Alvin. The H-bomb search did develop the methodology for computing SEPs (later called LEPs, for “local effectiveness probability”), but Richardson could not get an article about using probability to find the H-bomb published in an academic journal. The H-bomb hunt was a striking demonstration of how difficult it would be to win operational support for Bayes’ rule, even when something as tangible and terrifying as a missing thermonuclear bomb was concerned.

Still, although Bayesian updating was not used at Palomares, the success of the search strengthened Craven’s faith in scientific searches and the potential of Bayes’ rule. He and his team had learned how to compute subjective presearch hypotheses and weight their importance. They realized that the future of Bayesian search methods depended crucially on computer power and the portability of computerized information. This was not an insignificant realization. Richardson had been the only member of his graduate class in pure mathematics to take a computer course, and computer computation was still thought of as cowardly. Within months, though, Wagner, Associates acquired a punched-tape terminal, its first direct access to electronic computation. The next time they were called, the Bayesians would have better tools.

The navy got a dramatic opportunity to use Bayes' rule two years later, in the spring of 1968, when two attack submarines, one Soviet and the other U.S., disappeared with their crews within weeks of each other. As head of the Deep Submergence Systems Project, Craven was responsible for the search for both subs. Despite Bayes' limited role in the H-bomb search, both Craven and Richardson remained convinced the method was scientifically valid.

The first submarine to disappear was a diesel-fueled and missile-armed Soviet K-129, the source of Tom Clancy's fictionalized bestseller *The Hunt for Red October*. The U.S. Navy was alerted to its loss by a massive Soviet search in the Pacific off the Kamchatka peninsula along a major route frequented by its submarines. About the same time, U.S. underwater sensors recorded a "good-sized bang." The noise was far less than the sound of a sub imploding upon itself, but it occurred at a curious place, far from the Soviet search operation and on the International Date Line, at 40 degrees north and precisely 180 degrees longitude. Because the date line is a human artifact, the noise suggested a man-made event. Craven, one of a handful of U.S. personnel who knew of the "extremely classified" affair, hired Wagner, Associates for a full-scale probability analysis without ever telling them what they were looking for. Forty years later, Richardson still did not know he had worked on the search for the Soviet submarine.

Craven could think of only three plausible scenarios for the K-129's disappearance: "First, that the sound had nothing to do with the lost submarine. Second, that the sound was made by the submarine but that it did not sink and like Jules Verne's *Nautilus* was still gliding beneath the sea." Third, that the sub's watertight compartments were open when the crisis occurred and the vessel flooded so fast it did not collapse. Craven reasoned that if the sound recorded on the International Date Line came from the submarine, "then it was indeed not where it was supposed to be, which was why the Soviets could not find it."

Johnson, distracted during the tumultuous last months of his presidency, authorized a search for the Russian sub on the hypothesis that it might be a rogue, even though the prospect of finding it was slim. Eventually Craven concluded that the sub—armed with ballistic missiles and crewed by about 100 people—was indeed "a rogue, off on its own, in grave disobedience of its orders . . . [and possibly planning to attack Hawaii]. Since the Soviets didn't know how far off course their sub had been, *the Soviets would have had no idea that their ship was a rogue unless we told them.*"¹² U.S. authorities informed the Soviet leader, Leonid Brezhnev, of the bang's location, and in the face of evidence that his military might be out of control he could see détente as an attrac-

tive option. Later, Americans photographed the K-129 but were unable to raise it.

In May 1968, a few weeks after the Soviet sub sank, the U.S.S. *Scorpion*, a nuclear-powered attack sub, disappeared with its crew of 99 in the Atlantic Ocean. The *Scorpion* was cruising west, toward home, somewhere along a 3,000-mile submarine route between Spain and the East Coast of the United States. It was reportedly armed with two nuclear torpedoes. According to a study made in 1989, *Scorpion's* reactor and torpedos would be among at least eight nuclear reactors and 50 nuclear warheads to have been lost at sea; of these, 43 were on sunken Soviet submarines and eight originated with U.S. military activities. With the *Scorpion's* final resting place unknown, the military launched a full-scale search.

Craven and Andrews, by now the world's leading search experts, rapidly reassembled their H-bomb search crew. At first, the hunt stretched across the Atlantic Ocean. After some bureaucratic sleuthing, though, Craven learned that an ultrasecret listening post for "an unnamed agency" had recorded mysterious "blips" in extremely deep water about 400 miles southwest of the Azores. The location of the blips corresponded with the sub's expected itinerary and drastically narrowed the search area from a 3,000-mile-long rectangle to three or four square miles. Thanks to Craven, the investigation took a spectacular leap forward.

Craven organized a full-blown Bayesian hunt for *Scorpion* from the very beginning. When the H-bomb was lost off the Spanish coast, Craven had turned almost accidentally to Bayes in the hope of deflecting Congressional displeasure in case of failure. This time, the navy moved tentatively but with growing faith to exploit the method.

"Craven had confidence in the scientific approach from the very beginning but, to put it mildly, that wasn't everybody's thing," Richardson said. "Tough" is the word most often used to describe Craven, and for the next five months, from June to October 1968, he staunchly defended Bayes against skeptics. Although the H-bomb search in Palomares had failed to combine Bayesian priors and SEPs, Craven was enthusiastic when Richardson proposed doing so this time. A powerful computer in the United States would compute the probabilities of the various presearch hypotheses. Then this prior was to be combined and updated on shipboard with daily search results.

Shortly after the sub disappeared, Richardson was flown to the Azores to observe the surface search for *Scorpion* and to visit the USNS *Mizar*, a research ship conducting underwater operations. Personnel from the Naval Research

Laboratory, the Navy Oceanographic Office, and various equipment makers were aboard the *Mizar*, working 12-hour shifts around the clock. Over the next five months, they cruised through the area for weeks at a time, dragging across the ocean bottom a sledlike platform covered with a wide-angle camera, sonars, and magnetometers. The chief scientist on board the *Mizar*, Chester L. “Buck” Buchanan, had originally designed the equipment to find the *Thresher* and had improved it greatly since. He vowed not to shave until the *Scorpion* was found.

Scorpion searchers faced even more uncertainties than the H-bomb hunters had along the Mediterranean coast: a remote location 400 miles from land-based navigation systems, an ocean floor two miles down, and no eyewitness accounts pinpointing the *Scorpion*’s location. Navigational systems also introduced large errors and uncertainties. Two land-based radio networks, Loran and the new global Omega, were too imprecise to be useful, satellite fixes were available only irregularly, and transponders anchored to the ocean bottom were often indistinguishable from one another.

When Richardson arrived on board the *Mizar*, he found the ship following orders from Washington to search off Point Oscar—over and over again. Craven’s early analysis of acoustic data suggested that the *Scorpion* might have settled near Oscar. Using Bayes, however, Richardson tried to show graphically that they had oversearched Point Oscar and that very little probability remained of finding *Scorpion* there. Despite his brilliant demonstration, the search around Oscar continued. Washington would have to issue orders to change operations, and this would require persuasion based upon calculation of a detailed probability map, that is, a Bayesian prior.

“In all the operations that I’ve ever operated in you have strong personalities with their own ideas, and you have to argue—unless somebody [like Craven] in Washington shoves it down their throat,” Richardson said. “Otherwise, you have to convince people. And they have to come to their own conclusions that it’s the right way to do it.” Besieged by Craven, authorities in Washington later ordered that the prior probability map be treated as an important factor in the search.

On July 18, 1968, a month after the *Scorpion*’s disappearance, Craven gave “a brain dump” for Richardson and a new Wagner employee, Lawrence D. (“Larry”) Stone. Craven reported everything he had learned from his experts, and Captain Andrews presented a submariner’s view of what a sub might do under various circumstances. Working in Washington, Craven and Andrews outlined nine scenarios that might explain how *Scorpion* sank. Then

they assigned a weight to each according to how believable it was. This was the same approach Craven had used in the H-bomb search. Each scenario simulated *Scorpion's* movements and multiple uncertainties as to its course, speed, and position at the time of the blip.

One high-priority scenario was based on a mysterious piece of bent metal found by the *Mizar* during a quick survey of the region before the start of systematic searching. The metal was so shiny that it could not have lain very long on the sea bottom, and it was far away from the overly investigated Point Oscar.

Richardson and Stone carried their copious notes to Wagner, Associates' headquarters to quantify Craven and Andrews's assumptions and compute a prior "probability map" of the sub's location on the ocean floor. First, they established a search grid around the blip that Craven had identified as the probable location of the *Scorpion's* explosion. Each cell in the grid measured one mile north-south and 0.84 miles east-west, for a total of 140 square miles.

At Richardson's suggestion, the stateside search team made a key decision to use Monte Carlo methods to model the sub's movements before and after the accident. Physicists on the Manhattan Project had pioneered Monte Carlo techniques for tracking the probable paths of neutrons in a chain reaction explosion. Richardson substituted "little hypothetical submarines" for the neutrons. Academic Bayesians would not adopt Monte Carlo methods for another 20 years.

Starting with Craven's probable location of the explosion (the blip), a mainframe computed the probabilities that, in its death throes, the submarine changed course and traveled, for example, another mile in any of several random directions. Using Thomas Bayes' simplification, Richardson began by considering each of those directions as equally probable. Then, making a point at each new possible location, the computer repeated the process to produce new points, reiterating the procedure 10,000 times to make 10,000 points on the seafloor where the sub might have settled.

The use of Monte Carlo simulation to generate numbers based on Craven's presearch scenarios and weighting represented a big advance in search work. According to Richardson, "The nice thing with Monte Carlo is that you play a game of let's pretend, like this: first of all there are ten scenarios with different probabilities, so let's first pick a probability. The dice in this case is a random number generator in the computer. You roll the dice and pick a scenario to work with. Then you roll the dice for a certain speed, and you roll the dice again to see what direction it took. The last thing is that it

collided with the bottom at an unknown time so you roll dice for the unknown time. So now you have speed, direction, starting point, time. Given them all, I know precisely where it [could have] hit the bottom. You have the computer put a point there. Rolling dice, I come up with different factors for each scenario. If I had enough patience, I could do it with pencil and paper. We calculated ten thousand points. So you have ten thousand points on the bottom of the ocean that represent equally likely positions of the sub. Then you draw a grid, count the points in each cell of the grid, saying that 10% of the points fall in this cell, 1% in that cell, and those percentages are what you use for probabilities for the prior for the individual distributions.”

The 10,000 points were calculated on a mainframe computer in a small Princeton company that encoded classified data and punched it into paper tape. Such computers were available only on the mainland in the 1960s. Still in the future were so-called portable modems, 45-pound backbreakers for dialing into phone lines.

As cumbersome as it seems today, the time-shared mainframe made Bayes’ repetitive calculations feasible. It calculated the coordinates of *Scorpion*’s 10,000 possible locations and then counted the number of points that fell in each cell of the search grid. Lacking any sort of display screens, the computer printed out the numbers on ordinary teletype paper tape. Then the data were relayed over insecure public telephone lines to Richardson and Stone in Paoli. It was the only practical way of incorporating all the presearch data accumulated by Craven and Andrews in Washington into a detailed probability map.

Richardson later felt guilty about calculating only a “skimpy” 10,000 points, but at the time it seemed like a big number. Today’s computers refine detail even in low-probability areas. When the 10,000-point map was finished, it described the initial probabilities in 172 cells covering 140 square miles. Two cells, E5 and B7, stood out like rock stars under spotlights. With presearch simulation “hits” of 1,250 and 1,096, respectively, they were by far the most likely resting places for *Scorpion* and its crew. The next 18 most likely cells had far lower probabilities, between 100 and 1,000; and most cells (which had scores below 100) seemed almost irrelevant. The map was based on hours of conversations with Craven and Andrews, their scenarios, and their weighting. Unlike the H-bomb analyses of two years before, this map represented a real scientific advance, primarily because of the Monte Carlo calculations of the *Scorpion*’s possible movements.

By late July, the map was ready, and Washington ordered that it be

treated as an important factor in the search. It was now time for Bayesian updating, with data on the effectiveness of the fleet's search effort in each cell.

Aboard the *Mizar*, mathematicians from Wagner, Associates accumulated and recorded the effectiveness of each day's search. Stone and, later, two young students—Steven G. Simpson, a mathematics Ph.D. candidate from MIT, and James A. Rosenberg, a Drexel University undergraduate co-op student—worked on the area identified by Craven's scenarios about 2,000 yards from Buchanan's shiny piece of metal. Calculating by hand, they estimated the capabilities of the fleet's cameras, sonars, and magnetometers and combined them into a single number expressing the effectiveness of the search conducted in each cell of the sea-bottom grid. These numbers would eventually become the second component in Bayes' formula. Each morning the students had the unenviable task of tactfully advising a succession of naval commodores on the effectiveness of their searches: "Well, sir, I think it would be better if you did this rather than that."

Psychologically, a search can be difficult. Until the target is found, every day represents a failure. As Stone put it, "Bayes says that the longer you search without finding the target, the worse your prospects are, because the remaining time to detect the target gets longer, not shorter."¹³ On the other hand, those with confidence in Bayes' rule could trace their progress. "Areas that you search go down in probability," Richardson explained, "and areas that you haven't searched go up. So your updated probabilities get higher where you haven't been looking. . . . And generally, it's always most optimal to keep looking in the highest probability area. The next day you have a high probability area somewhere else, probably not where you searched, and they pop up somewhere else the third day, and you just keep doing it and doing it and doing it. And unless you've made some drastic mistake, you'll eventually find what you're looking for."

As it was during the H-bomb hunt, the biggest problem turned out to be overestimates of the sensors' capabilities. Most of them had never been tested or evaluated systematically for how well they could detect a piece of metal to the left or right of their detectors. Thinking over the problem, Richardson realized, "So you had two uncertainties and, if your objective is to come up with the mathematical expression for how to allocate resources optimally, that's an interesting point."

As naval commanders came and went during the *Mizar*'s five cruises to the search site, Bayes' rule became the group memory of the search and its coordinating principle. For the first time, the rule was used from beginning

to end of a long search. Unfortunately, not everyone saw the Monte Carlo prior map as a powerful tool for directing Mizar's search. It took almost a month to get the map to the scene of operations for its Bayesian updating. Communications between land and sea were so poor that Stone finally hand-carried the map to the Azores on August 12. Not until the research ship's fourth and fifth cruises in October—five months after *Scorpion* disappeared—did the detailed prior distributions based on Craven's and Andrews' scenarios become available.

The Mizar's fifth and last cruise was originally planned to test the sensors, refine the underwater tracking system, and study the contours of the ocean bottom. By this time Craven had organized acoustical studies to more precisely calibrate the location of the blip recorded by the supersecret sensors. Small depth charges were exploded in the ocean at precisely known positions, and their sounds were used to refine the information recorded by naval listening posts during the *Scorpion*'s last moments. Every day Craven's acoustical analyses edged the most likely spot for the *Scorpion* closer to Buchanan's shining piece of metal.

In late October the increasingly impatient and by now heavily bearded Buchanan finally got approval to investigate the shiny metal. As Mizar's sled made its 74th run over the ocean floor, its magnetometer spiked high at several anomalies in cell F6. Returning to the area on October 28, Mizar struggled to pinpoint the spot again. Finally its cameras revealed, lying on the sea bottom, partially buried in sand, the submarine *Scorpion*. A poorly functioning sonar detector had previously passed right over the sub without finding it. Word that Buchanan was shaving his beard spread rapidly to the United States.

Richardson was back in the States when he got the phone call. "They gave me the location in code," he said, "and I plotted it up, and at first I thought it was going to plot right smack in the middle of that high probability square, and I was really excited." Instead, it was 260 yards away, close to the mysterious piece of shiny metal found at the beginning of the search. The scrap was later determined to be a *Scorpion* fragment. Still, Richardson joked ruefully, 260 yards off in a 140-square mile area of open sea was "close enough for government work."

Years later, Captain Andrews argued that Bayes was only one day and a half mile behind Buchanan. If Buchanan had not returned the Mizar to the shiny metal that day, Craven's presearch probabilities, updated with his later acoustical studies, would have found *Scorpion* first.

On November 1, five months after the search began, Rosenberg, the co-op student from Drexel, hand-carried the photos of *Scorpion* back to the United States. Excluding time spent studying a misleadingly magnetic, hull-shaped rock, the search sled had located the submarine after scanning 1,026 miles of ocean bottom at a speed of one knot for the equivalent of 43 days, two days earlier than Bayesian predictions.

President Johnson was told “the highest probability was that the sinking was caused by an accident on board the submarine.”¹⁴ This time he may have listened to probabilities.

Analyses of the sounds made by the *Scorpion* suggested that it had been traveling east, rather than west, when it sank. Twenty years later Craven learned that the sub could have been destroyed by a “hot-running torpedo.” Other subs in the fleet had replaced their defective torpedo batteries, but the navy wanted *Scorpion* to complete its mission first. If *Scorpion* had fired a defective torpedo, it would have missed its target and probably turned back and struck the sub that had launched it.

Anxious to document the methods used in the search for *Scorpion*, the Office of Naval Research commissioned Stone to write *Theory of Optimal Search*. Published in 1975, it is an unabashedly Bayesian book incorporating applied mathematics, statistics, operations research, optimization theory, and computer programs. Cheaper and more powerful computers were transforming Bayesian searches from mathematical and analytical problems to algorithms for software programs. Stone’s book became a classic, important for the military, the Coast Guard, fishermen, police, oil explorers, and others.

While Stone was writing his book, the United States agreed to help Egypt clear the Suez Canal of unexploded ammunition from the Yom Kippur war with Israel in 1973. The explosives made dredging dangerous. Using the SEPs developed in Palomares, it was possible to measure the search effectiveness to get the probability that, if a bomb had been there, it would have been spotted. But how could anyone estimate the number of bombs remaining in the canal when no one knew how many were there to begin with? Wagner, Associates chose three priors with different probability distributions to express high, middle, and low numbers. Next, using the handy system of conjugate priors described by Raiffa and Schlaifer in 1961, they declared that each prior would have a posterior with the same class of probability distributions. This produced three tractable distributions (Poisson, binomial, and negative binomial) complete with those statistical desiderata, mean values and standard deviations. Computing became “a piece of cake,” Richardson reported, but

it proved impossible to explain the system to hardened ordinance-disposal specialists with missing fingers. In the end, no one talked about Bayes at Suez.

Up to this point, postwar Bayes had searched only for stationary objects like bombs in a canal, or H-bombs and submarines on the ocean floor. Technically, these were simple problems. But shortly after the Suez Canal was cleared and *Theory of Optimal Search* was published, intensive efforts were made to adapt Bayesian methods to moving targets: civilian boats adrift in predictable currents and winds.

The technology was a perfect match for U.S. Coast Guard rescue coordinators like Joseph Discenza, whose job in the late 1960s was to answer the telephone when someone called saying, "My husband went out fishing with my son, and they're not back."¹⁵ After checking area ports of call for the boat, he used a Coast Guard Search and Rescue Manual to estimate by hand the target's location and its probable drift.

"Like a dog with a bone in his teeth," Discenza started to computerize the Coast Guard's manual.¹⁶ He studied search theory and earned a master's degree at the Naval Postgraduate School in Monterey, California, and a Ph.D. at New York University. Along the way Discenza discovered that ever since the Second World War the Coast Guard had been using the Bayesian search theory developed by Koopman to find U-boats in the open ocean. Discenza filled in the corporate memory gap between the 1940s and the 1970s. "The Coast Guard was very Bayesian. Even when they're doing it manually, they're doing Bayes," Stone said. But until Discenza, they were like early casualty actuaries, using Bayes' rule without realizing it.

Joining forces with Discenza, Wagner's company designed a computerized search system based on Bayesian principles for the Coast Guard. A natural outgrowth of the H-bomb and *Scorpion* searches, it combined clues about a vessel's original location and subsequent movements into a series of self-consistent scenarios and then weighted them as to their likelihood.

The Coast Guard ruled that estimating probabilities and weights should be a group decision. Each individual involved should weight the scenarios privately before they were averaged or combined by consensus. Above all, no scenario should be discarded. "To leave out subjective information is to throw away valuable information because there is no unique or 'scientific' way to quantify it," Stone urged.¹⁷

What if a ship in distress radioed its position but a small plane reported seeing it an hour later a hundred miles away? One or the other had made

an error in position, but neither report should be ignored; both should be assigned relative reliabilities. As Stone commented, "Discarding one of the pieces of information is in effect making the subjective judgment that its weight is zero and the other weight is one."

Bayesian updating and, at Richardson's insistence, Monte Carlo techniques were incorporated into the Coast Guard system in 1972, almost two decades before university theorists popularized the method or the term "filters." The Monte Carlo methods estimated an enormous number of possible latitudes, longitudes, velocities, times, and weights for each lost ship to pinpoint 10,000 possible target locations.

Stone also used a Bayesian procedure, an early version of a Kalman filter, to separate and concentrate the data or signals according to specified criteria and to weigh each possible path of a target's motion according to its believability. The technique did not become popular among academics until the 1990s, but it saved military and space contractors immense amounts of time in the 1960s because their computers had little memory or power. Before Rudolf E. Kalman and Richard Bucy invented the procedure in 1961, each original observation had to be completely recalculated every time a new one appeared; with the filter, new observations could be added without having to do everything all over again. Kalman vehemently denied that Bayes' theorem had anything to do with his invention, but Masanao Aoki proved mathematically in 1967 that it can be derived directly from Bayes' rule. Today, it is known as a Kalman or a Kalman-Bucy filter.

Once Monte Carlo methods and filters were adopted, even untenable and highly improbable paths produced valuable information and helped searchers determine which of the remaining paths were more likely. As more information arrived from sensors, Coast Guard aircraft, weather reports, tide tables, and charts of prevailing currents and winds, the data were converted into likelihood functions and then combined with priors about the target's movements to predict its probable location. As data accumulated with each iteration, the filter concentrated a relatively small number of highly probable paths.

The Coast Guard's system was up and running in 1974 when a tuna boat sank off Long Beach, California. Two days later, purely by chance, a freighter found 12 of its survivors in a lifeboat. Using their new technology, the Coast Guard calculated backward from the chance rescue to the tuna boat's probable capsizing point and then forward again using probability maps of ocean currents and Bayesian updating. Armed with a Bayesian probability map, the Coast Guard rescued three more men the next day. Another successful search took

place two years later after a ship capsized and sank while crossing the Pacific. Five sailors took to sea in two life rafts. Twenty-two days later, also by chance, two survivors were found in one of the rafts. Six days after that, the same Coast Guard program found a third survivor, who had been adrift for 28 days.

Bayes had found stationary objects lodged on the seafloor and had tracked boats drifting with predictable ocean currents and the wind. But what about locating and following evasive prey, a Soviet submarine, say, or a moving target operated by human beings? Could Bayes accommodate human behavior?

"It's the Cold War, and there are submarines out there that are a threat to the U.S.," recalled Richardson. "They're moving targets, so why not do something in antisubmarine warfare. . . . It started in the 1970s and continued for two decades, and I personally did a lot of work searching for subs in the Atlantic Ocean and the Mediterranean."

When the future vice admiral John "Nick" Nicholson took command of the U.S. submarine fleet in the Mediterranean in 1975, he secured a \$100,000 grant from the ONR to bring Richardson to Naples for a year. It was one of ONR's biggest contracts, and navy accountants considered it a waste of money. But the Mediterranean was full of Soviet and NATO ships and submarines eyeing one another; the Soviets alone had 50 vessels, including ten submarines. By the early 1970s the "Med" was so crowded that the U.S. and Soviet governments signed a pact to reduce collisions. When the U.S. Navy began routine tracking of Soviet subs in the Mediterranean in 1976, Nicholson thought Richardson and Bayes' rule could help.

Starting from scratch, Richardson cranked intelligence information into an antiquated computer in Naples: previous submarine tracks; particular types of Soviet sub that were apt to take a particular route and perform certain maneuvers; and reports from sonobuoys, passive acoustic listening devices that were dropped by aircraft into fixed underwater surveillance systems. Unlike Koopman's purely objective analysis of radio transmissions and submarine tracks during the Second World War, Richardson was making subjective assessments of the behavior of Soviet officers. He was also using real-time feedback of actual search results, something that submarine hunters in the Second World War would have regarded as science fiction. To all this intelligence data Richardson added the islands, sea mounts, and tight passages in the region's constricted geography. These natural obstacles became surprisingly helpful features.

By definition, tracking involves uncertainties and estimations that are

far from ideal. Parameters change as new data appear, and “to make matters worse, the data can be remarkably uninformative and obtained from a number of different sources,” as Stone wrote.¹⁸ An optical scanner might spot a distant periscope emerging a foot above the horizon for 10 seconds but fail to identify it as a submarine. Operators watching radar signals on their computer screens could not always distinguish a sub from a surface ship. Arrays of acoustic hydrophones extended over the seafloor for hundreds of miles to detect low-frequency acoustic signals emitted by submarines, but their data were often highly ambiguous. Different targets, for example, radiated acoustic signals at the same or nearly the same frequency. Even the ocean distorted sounds. Sound waves bend with every change in water temperature, and the roar of breaking waves affects noise-to-signal ratios. Bayes’ common currency—probabilities—fused information gathered from these various sources. Amid such vague and ephemeral reports, Bayes’ rule was in its element.

One summer day in 1976 a Soviet nuclear-powered submarine slipped through the Strait of Gibraltar and entered the Mediterranean. It was a 5,600-ton Echo II class vessel, armed with cruise missiles that could be fired from the surface. The U.S. fleet tracked it as far as Italy before losing it. No one could tell when it would pass through the Sicily Straits into the eastern Mediterranean.

Besides the submarines under his command, Nicholson had been assigned four antisubmarine destroyers that pulled experimental sleds packed with trailing-wire sonar detectors. Arranging his forces across the Sicily Straits so the destroyers would have a chance to detect the Soviet sub passing through, Nicholson waited tensely. “The wait went on longer and longer than all of our operations [intelligence] people were expecting,” Nicholson related years later. “Tony kept working his program and he kept saying, ‘I still think there’s an x percent possibility that it hasn’t gone through yet.’”¹⁹

Nicholson’s superiors were pressuring him to move his submarines and surface ships over to the eastern Mediterranean to look for the sub there. But he was an old hand at pressure; he had been executive officer and navigator of the second nuclear submarine to go under the Arctic ice cap to the North Pole and had commanded the first nuclear sub to go from the Pacific to the North Pole in the winter.

Richardson, still poring over the old computer, urged Nicholson to ignore his commander. “I think we should take at least one to two more days,” he said. He estimated the probability that the sub had not yet slipped through

the straits at about 55%. Nicholson didn't know how much confidence to place in Richardson's system. It was new, it included subjective assessments of human behavior, and it was being used for real-time decision making. But Richardson was filling a void other intelligence and operations experts could not. Making a bold decision that could have destroyed his career, Nicholson decided to wait. "And, lo and behold, we made contact and were able to track the guy through the Strait." The Sixth Fleet was jubilant, and, as Richardson described the reaction of the brass, "most everyone became a believer" in Bayesian search methods.

Thanks to the trailing-wire sonar detectors, every time the Soviet submarine came to the surface in the eastern Mediterranean one of Nicholson's destroyers was cruising nearby. Their skippers were under orders not to come too close to the sub, but, as Nicholson says, "These destroyer guys don't listen very well."

One rather clear Sunday morning the Soviet submarine surfaced with its sail (a metallic structure covering periscopes and masts) about four feet out of the water. Waiting nearby was one of Nicholson's destroyers, the 3,400-ton *Voge*. To everyone's surprise, the Soviet sub turned toward the *Voge* and charged at full speed.

As Nicholson recounts the story, "Everybody on the ship was taking pictures of this thing, the submarine moving along at 20 knots with its sail out of the water, when the skipper of the surface ship slowed for some reason. The submarine skipper apparently didn't keep his eye right on it, and the first thing you know, the submarine rammed right into the *Voge*. We believe the Soviet captain was trying to cut the trailing-wire sonar that had given him such fits."

The sub was badly damaged, and its captain was relieved of his command that same night. The *Voge* was towed to France for repairs. The incident proved the value of Richardson's tracking and of trailing-wire sonar systems on surface ships. Later, Bayesian methods tracked Soviet submarines in the Atlantic and Pacific, although after the breakup of the USSR in 1991 Russian out-of-area submarine deployments were greatly reduced.

"The antisub warfare work was pretty much the highlight of things that were really Bayesian," Richardson reflected. "... It was like being back in Spain again. I was ten or fifteen years older, sitting up all night, running my computer and briefing the admiral in the morning. ... That's kind of the ultimate happiness, when you can make things move around in the world based on your ideas."

Meanwhile, the military that had been so slow to embrace Bayesian search theory was exploring its use for identifying asteroids speeding toward Earth and for locating Soviet satellites as they orbited in space. In 1979 NATO held a symposium in Portugal to encourage the solution of “real problems” with Bayesian methods.

Most of the attendees were from the military, but Richardson and Stone gave talks about their submarine hunts, while others spoke about search and rescue and oil deposit exploration. Among the civilian attendees was Ray Hilborn, a newly minted zoology Ph.D. who was interested in saving the fish populations in the world’s oceans. He had gotten his first exposure to simple Bayesian applications at the East–West think tank run by Raiffa in Vienna six years earlier.

Hilborn was struck by the fact that people at the NATO conference dealt with practical problems that required making decisions. His own job involved setting legal limits on fishing for particular species, and, listening to the speeches, he said to himself, “God, this really is the way to ask the questions I want to ask. Everyone who is actually involved in the real world does things in a Bayesian way. The limit of [frequentist] approaches just isn’t obvious until you actually have to make some decisions. You have to be able to ask, ‘What are the alternative states of nature, and how much do I believe they’re true?’ [Frequentists] can’t ask that question. Bayesians, on the other hand, can compare hypotheses.”²⁰ It would take him almost 10 years to find a fisheries problem for Bayes, but Hilborn was a patient man.

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